# What Should Reference Labels Buy? Estimation, Certification, and Repair in Conjunctive Evaluations

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**Abstract**

Checking an LLM judge against reference labels can improve an accuracy estimate, certify an outcome, or repair an existing verdict. These gains need not coincide when every rubric condition must pass. We study constructed all-pass targets on RuVerBench and JudgmentBench, comparing update rules on identical queried labels and allocations at equal query budgets. Eager updating can expose a false pass while correcting individual labels; waiting for a certificate prevents that error but delays repairs. The delay depends on the gap between clearing mistaken rejections and checking the whole rubric. In the coding frame, certificate-gated updating leaves 5.3 more disagreements per 100 tasks at a 20% criterion budget. A substantial legal-frame effect occurs under a strongly FAIL-dominated judge and persists in direction under separate reference and task-composition checks. A fixed certification-plus-sampling allocation increases certification in all nine analysis cells, while its repair gains depend on which verdicts receive the certificates and its accuracy estimates become less precise. Finally, equal criterion confusion counts can favor opposite update rules, including in a constructed example with unchanged task verdicts. Together, the results identify distinct information requirements for estimation, certification, and repair. Design-based estimators and finite-population bounds distinguish sampling precision from conclusions determined by observed labels.

## 1 Introduction

Rubric-based evaluation breaks a complex output into several judgments: whether it follows the requested format, supplies the necessary evidence, and avoids specified errors. When success requires every condition to hold, one failed condition determines the verdict. APEX-Agents uses this rule for task-level Pass@1 (Vidgen et al., 2026). We study the consequences of reference checking under the same logical rule, using task-linked labels from RuVerBench and JudgmentBench. We construct all-pass targets from these releases; their native studies use different headline metrics. Full JudgmentBench includes both positive requirements and zero-occurrence penalty conditions. It provides a FAIL-dominated stress test, alongside a binary-only target and coding and research frames with different error distributions.

Partial checking can repair the individual labels and still damage a verdict. In our full JudgmentBench target, checking all 4,316 criteria on which GPT-5.4 and GPT-5.4-mini disagree corrects 1,217 criterion errors. It repairs four verdicts but turns five correct rejections into false passes, changing task disagreements from 244 to 245. The net change is small; the important feature is its path. The checks remove mistaken reasons for rejection before reaching the conditions that actually fail. Waiting for a complete certificate prevents such a change, but also postpones repairs when the initial rejection was wrong.

A limited checking budget introduces a second problem. The same reference label can contribute to estimating the judge's overall accuracy, certify a particular task, or change an existing verdict. A checked failure may certify a task that was already correctly rejected. Correcting a mistaken rejection may repair another task before its remaining conditions are checked. These are distinct uses of evidence. An allocation that produces more certificates can therefore leave fewer, the same number, or more incorrect verdicts than another

allocation at the same cost.

Prior work supplies both the checking tools and the warning that correction can introduce errors. Multi-predicate screening and Boolean-function evaluation study costly condition checks (Krivosheev et al., 2018a; Gkenosis et al., 2022); ActiveClean separates selection from updating (Krishnan et al., 2016); correction-and-corruption analysis accounts for opposing outcome changes (Reitich, 2026); and Query-Guided Verification studies local verification errors propagated through AND/OR queries (Schreiber and Amsterdamer, 2026). Our question concerns an existing rubric verdict as reference labels arrive. We separate *acquisition*, which chooses those labels, from *updating*, which changes the verdict. *Eager updating* substitutes each queried label and recomputes the conjunction. *Certificate-gated updating* waits until the queried labels determine the outcome. Applying both rules to the same queried set isolates the effect of waiting for evidence, while an equal-budget allocation comparison measures what different query sets accomplish.

The study makes three contributions. First, it measures the competing effects of prevented false passes and delayed repairs over the query budget, with exact event probabilities and separate reference and task-composition checks. Substantial delay also appears in coding data without JudgmentBench's penalty items, while task-completing acquisition sharply limits the difference between update rules. Second, it explains allocation outcomes through the placement of certificates on initially correct or incorrect verdicts and measures the accompanying estimation cost. Third, it exhibits the information lost in criterion summaries: compatible label arrangements can favor opposite updates, while trusted summaries and linked labels determine different task-level quantities. The accounting identities organize these comparisons; the empirical contribution is the measured dependence on the error configuration, query structure, scoring target, and reference.

The paper addresses three questions:

**RQ1.** For the same queried reference labels, when does certificate-gated updating prevent more errors than it delays repairs, and how do error configurations and query order shape that balance over the budget?

**RQ2.** How do sampling units and query allocations trade estimation precision against certification coverage and verdict repair, and when do certification gains translate into fewer residual disagreements?

**RQ3.** Which task-level conclusions are determined by criterion summaries and queried reference labels, what additional task associations are needed, and how does identification differ from statistical estimation?

Our evidence is a retrospective replay of released labels and saved judge predictions. We use those labels as specified references, and separately examine how changing the human reference changes the findings. This permits controlled comparisons without treating every recorded disagreement as an adjudicated judge error.

# 2 Setting and study design

## 2.1 Four evaluation targets

An evaluation frame contains $N$ units. Unit $i$ has $k_i$ retained criteria, with binary reference labels $Y_{ij}$ and judge predictions $P_{ij}$; one denotes PASS. Write $M = \sum_i k_i$, $Z_i = \prod_j Y_{ij}$, and $\widehat{Z}_i = \prod_j P_{ij}$. A unit is a task in RuVerBench and an output-by-rater annotation in the original JudgmentBench frame.

Throughout the results, a *verdict* belongs to this evaluation unit. Original JudgmentBench counts weight annotations, including repeated ratings of the same output. The matched-reference analysis instead uses one unit per output; we name that change whenever reporting its effects. Base tasks are the clusters used for task-composition resampling.

We distinguish criterion microaccuracy, $a = M^{-1} \sum_{ij} \mathbf{1}(P_{ij} = Y_{ij})$; the aggregate task pass rate, $\theta = N^{-1} \sum_i Z_i$; individual certification, which determines a particular $Z_i$; and verdict repair, which reduces disagreement between reported verdicts and $Z_i$. The first two are population quantities, while the last two concern individual outcomes. Counting erroneous criteria found by a policy is a further diagnostic, not a substitute for any of these targets.

A query reveals one reference bit. The query controller sees predictions, metadata, and previously queried labels; it cannot inspect future references. Unless additional information is explicitly supplied, unqueried bits can take either value. In this query-only model, one observed FAIL certifies failure and observing every bit as PASS certifies success. Classical Boolean evaluation provides the foundations for these certificates (Gkenosis

**Table 1:** Finite analysis frames. $N$ counts tasks in RuVerBench and output-by-rater annotations in JudgmentBench; $M$ counts criterion records. FP and FF compare the initial judge conjunction with the selected reference. The binary-only rows reuse the JudgmentBench annotations.

| Frame / primary judge | $N$ | $M$ | Ref. PASS | FP | FF |
|---|---|---|---|---|---|
| DR Gemini 3.1 Pro | 284 | 1,615 | 28 | 8 | 2 |
| DR Qwen-plus | 284 | 1,615 | 28 | 28 | 3 |
| DR GPT-5.4 low | 284 | 1,615 | 28 | 8 | 2 |
| AC Qwen-plus | 210 | 843 | 117 | 31 | 10 |
| AC DeepSeek v4-pro | 210 | 843 | 117 | 14 | 16 |
| JB GPT-5.4, full | 1,539 | 23,487 | 227 | 24 | 220 |
| JB GPT-5.4-mini, full | 1,539 | 23,487 | 227 | 36 | 223 |
| JB GPT-5.4, binary-only | 1,539 | 20,229 | 266 | 181 | 67 |
| JB GPT-5.4-mini, binary-only | 1,539 | 20,229 | 266 | 457 | 36 |

et al., 2022; Ünlüyurt, 2004).

For a queried set $Q$, *eager* updating replaces $P_{ij}$ by $Y_{ij}$ on $Q$ and recomputes the conjunction. *Certificate-gated* updating retains $\widehat{Z}_i$ until $Q$ certifies an outcome, then reports that outcome. The gate therefore governs revisions; unresolved initial predictions remain visible. We count false passes (reported PASS, reference FAIL), false fails (reported FAIL, reference PASS), and their sum, denoted $E$.

## 2.2 Data and constructed scoring targets

RuVerBench supplies reference judgments on agentic coding (AC) and deep research (DR) criteria (Peng et al., 2026). We use saved predictions from four initial judge–domain runs and one additional GPT-5.4 low-reasoning DR run. Appendix G records their prompts, generation settings, parsing and technical recovery. The conjunction covers the criteria retained in the released frame after upstream filtering.

JudgmentBench provides rubric annotations for legal outputs and saved GPT-5.4 and GPT-5.4-mini scores (Yang et al., 2026). Its native study concerns quality assessment by rubrics and preferences. We construct a strict all-pass target: every positive binary item must receive full credit, and every penalty occurrence count must be zero. The binary-only sensitivity removes penalty items. These choices define different success events. The original frame contains 1,539 annotations on 1,314 outputs from 30 base tasks, with 49 human raters. Table 1 describes all nine analysis cells; cells sharing outputs are not independent replications.

The full JudgmentBench target exposes a pronounced error imbalance: GPT-5.4 predicts only 31 PASS verdicts, of which seven agree with a reference PASS. Removing penalties raises predicted passes to 380 and reference passes to 266. Of the 220 initial full-target false fails, 156 reject only penalty conditions and 64 reject both item types. This is a useful stress test for update delay. It also means that its effect sizes should be interpreted alongside the changed target and the other frames.

## 2.3 Queries, budgets, and sensitivity designs

The broader policy study compares seven acquisition rules: uniform criterion sampling; short circuit with natural, randomized, or judge-failure-first within-task order; disagreement-only checking; disagreement checking followed by random continuation; and disagreement-prioritized short circuit. A disagreement flag records any available auxiliary prediction that differs from the primary. Short-circuit policies stop a task after a queried FAIL or a completed PASS certificate. Our analytical budget comparison focuses on uniform criterion sampling and disagreement-first continuation. The latter visits flagged criteria task by task, then samples globally from the remaining criteria. Task order determines which disagreements are reached during the first phase.

We evaluate 101 budget caps from zero to the entire criterion frame, three task orders, and 31 randomized priority realizations, with a separate canonical seed-0 realization. Task orders are release order, ascending predicted-failure count, and random. A stopped policy can spend less than its cap, so comparisons report actual queries. The focal 20% allocation experiment compares criterion simple random sampling (SRS) with

a fixed mix: up to half the budget goes to judge-first short circuit, followed by independent SRS from the unqueried remainder. The split is held fixed throughout the comparison.

Two additional designs examine stability. First, 206 repeatedly rated JudgmentBench outputs receive symmetric reference panels A and B, selected by an ID-hash rule. Both judge prediction vectors are held fixed. Crossed acquisition/evaluation references separate changing which labels are queried from re-reading the same query positions under another reference. Second, 399 whole-base-task resamples retain complete output, rater, and criterion blocks. Each replicate recalculates frame size and budget and reruns acquisition. Query-order variation, reference variation, and task-composition variation are reported separately. Appendices B–F provide the formulas, sampling procedures, and reconstruction constraints.

# 3 RQ1: How updating changes task verdicts

## 3.1 A repair can expose another error

Consider a task with reference bits $(1, 0)$ and judge bits $(0, 1)$. Both conjunctions initially say FAIL. Checking the first criterion changes the eager bits to $(1, 1)$, producing a false PASS. Gating retains FAIL because the unchecked second criterion could still fail. The partial correction removed the reason for the original rejection without discovering the actual failure.

Now consider reference bits $(1, 1)$ with judge bits $(0, 1)$. Checking the first criterion repairs the eager verdict, but gating waits for the second check before reporting PASS. The same query structure that prevents an exposed false pass can postpone a valid repair. The empirical question is which of these events occurs more often at a given budget.

Let $A_b$ count newly introduced false passes under eager updating and $D_b$ count initial false fails that eager has repaired while gating still lacks a certificate.

**Proposition 1** (Update accounting)**.** *For a fixed reference and the same queried set under both update rules,*

$$E_b^{\text{gate}} - E_b^{\text{eager}} = D_b - A_b. \tag{1}$$

Partitioning tasks by their initial and reference verdicts proves the identity (Appendix A.3). The net effect depends on when erroneous rejections are corrected and actual failures are discovered. Their positions within each task and the query order determine the length of that interval.

For uniform sampling of $m$ criteria, an event requiring $r$ particular bits to be observed and $f$ other bits to remain unobserved has probability

$$h(M, m; r, f) = \frac{\binom{M-r-f}{m-r}}{\binom{M}{m}}, \tag{2}$$

with infeasible combinations assigned zero. For a susceptible initially correct FAIL, all predicted failures must be queried while every reference failure remains unqueried. For an initial false FAIL, delayed repair requires querying all predicted failures without completing the rubric. Summing these task events gives $\mathbb{E}[A_b]$ and $\mathbb{E}[D_b]$. Applying the same calculation within the active priority group gives exact expectations for disagreement-first continuation. The calculation is exact for the fixed reference and integrates over query randomization while retaining the observed dependence among criterion errors.

## 3.2 Repair delay across budgets and scoring targets

The balance changes substantially with budget and frame (Table 2). In full JudgmentBench at a 20% cap, preventing new false passes slightly outweighs delayed repair. At a 50% cap the expected net difference is already +44.8 annotation disagreements; it reaches +96.3 at 84%, where delay outweighs prevention by about nine to one. Both rules converge to the reference at complete observation. The maximum therefore describes an interior budget, rather than the final outcome of completing the audit.

The coding comparison shows that a substantial delay does not require JudgmentBench's penalty items or its strongly FAIL-dominated predictions. In AC DeepSeek, 117 of 210 tasks pass the reference target. At a

**Table 2:** Matched-query updating under disagreement-first continuation, release task order. Positive differences mean more disagreements under gating. Full and binary-only JB use 1,539 annotations; AC uses 210 tasks. Expectations integrate query randomization; the final column averages 31 saved orders. Entries are rounded independently.

| Frame / target | Cap | Queries | $\mathbb{E}[A_b]$ | $\mathbb{E}[D_b]$ | $\mathbb{E}[D_b - A_b]$ | Order mean |
|---|---|---|---|---|---|---|
| JB full | 20% | 4,697 | 7.1 | 5.2 | −2.0 | −2.1 |
| JB full | 84% | 19,729 | 11.9 | 108.2 | +96.3 | +97.6 |
| JB binary-only | 20% | 4,046 | 28.2 | 36.4 | +8.2 | +8.3 |
| AC DeepSeek | 20% | 169 | 0.9 | 12.1 | +11.2 | +11.2 |

20% budget, the expected difference is +11.15 disagreements, or 5.31 per 100 tasks. This is an existing-frame comparison; the separate reference and composition checks below cover JudgmentBench and DR.

The available repair backlog and its completion times explain the differences between frames. Full JudgmentBench has 220 initial false fails, compared with 67 under binary-only scoring and two in DR Gemini. At 84%, the 156 full-target false fails rejected only on penalty items contribute 77.9 of the 108.2 expected delayed repairs. This locates most delay in that group. Removing penalties changes the success event, rubric length, priorities, and budget together, so it does not isolate a causal effect of item type. Length alone also cannot determine delay: if every criterion is predicted FAIL, clearing all erroneous rejections completes the PASS certificate at the same time.

Acquisition order constrains the size of the update effect.

**Proposition 2** (Unresolved-task bound)**.** *Let $U(Q)$ count partly queried, uncertified units. With the same fixed reference and query set, $|E^{\text{gate}}(Q) - E^{\text{eager}}(Q)| \le U(Q)$. A serial policy that certifies each task before starting the next has $U(Q) \le 1$ at every prefix.*

Both rules agree on untouched and certified units, which proves the bound. Thus a large update effect requires checks dispersed across unfinished tasks. Figure 1 shows the complete-budget effects of disagreement-first continuation across three frames.

Task order matters while the disagreement tier is incomplete. In full JudgmentBench, that tier contains 18.4% of all criteria. Once it has been exhausted, every task order reaches the same unflagged pool and uses the same continuation design. The three orders therefore give identical expected endpoint effects after this point; their saved paired realizations agree as well. Appendix B.2 reports early- and later-budget comparisons, separating task ordering from randomized priorities within each order.

## 3.3 Reference and task-composition sensitivity

Task-composition resampling separates the small early advantage from the later repair delay. In full JudgmentBench, the 20% effect is −0.13 disagreements per 100 annotations and its pointwise composition range crosses zero. At 50%, the effect is +2.91 with range $[1.05, 5.20]$; at 84%, it is +6.26 with range $[3.51, 9.44]$. Every one of the 399 resamples gives a positive effect at both later budgets. For binary-only at 20%, the range is $[-0.44, 1.50]$: its original positive contrast is sensitive to task composition. These are 2.5–97.5% pointwise ranges from the specified resampling design.

The matched-output analysis confirms the high-budget delay while showing its dependence on the reference. On the 206-output panel, the 84% effect is 6.0 or 3.9 disagreements per 100 outputs under references A and B, respectively. Both references give a small positive effect at 20%. This panel uses one record per output and a fixed judge anchor, so its comparison with the original annotation frame also changes weighting and sample composition. Within the panel, switching the reference alone can reverse small contrasts; binary-only uniform sampling at 20% gives mean differences of +1 and −1 output disagreement.

Repeated ratings explain why reference sensitivity belongs beside the headline results. The repeated-rating subset contains 56 reference-PASS annotations: 54 initial false fails and two true passes. Among those 54 false fails, 32 have another rater who also rejects the output. Only 22 of the 56 reference-PASS annotations are supported by every available peer. These observations establish disagreement among references, without determining which rater is correct.

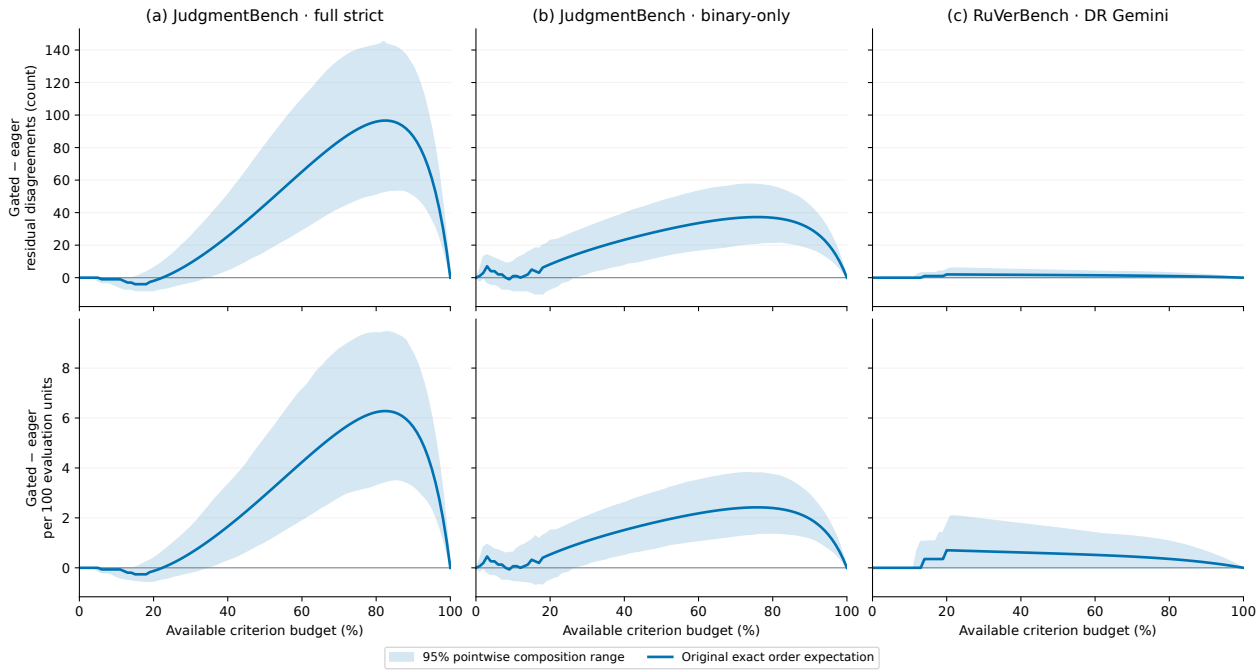



**Figure 1:** Complete-budget update comparisons in full JudgmentBench, binary-only JudgmentBench, and DR Gemini. The plotted disagreement-first continuation policy uses release task order. Original effects are analytical expectations; the shaded pointwise ranges summarize 399 whole-base-task resamples. Each replicate reruns the policy at its recalculated budget. These ranges describe task-composition sensitivity, separately from query-order or reference-panel variation. Both endpoint effects are exactly zero.

**Table 3:** Peer evidence for the 220 initial false fails in full JB GPT-5.4. Contributions to $\mathbb{E}[D_b]$ at the 84% cap use the original disagreement-first query design and are rounded independently. Peer status describes observed agreement, not reference correctness.

| Other raters on the same output | Initial FF | Contribution to $\mathbb{E}[D_b]$ | Share |
|---|---|---|---|
| At least one says FAIL | 32 | 15.5 | 14.3% |
| All say PASS | 22 | 8.7 | 8.0% |
| No other rater available | 166 | 84.1 | 77.7% |
| Total | 220 | 108.2 | 100% |

Most of the original expected delay comes from annotations with no repeated-rating evidence. Table 3 partitions it without changing the query design: about 14% has a peer FAIL, 8% has unanimous peer PASS support, and 78% has no peer rating. Appendix B.3 gives the per-unit calculation. The 32/54 disagreement proportion cannot be extrapolated to the unreplicated annotations.

# 4 RQ2: What additional certificates buy

## 4.1 A common-budget comparison

We compare three uses of the same budget: random criterion sampling, a fixed certification-plus-sampling mix, and pure judge-first certification (Table 4). The mix increases certification and widens accuracy intervals in both focal frames. Pure certification produces still more resolved verdicts and fewer residual disagreements, but its adaptive queries do not supply the probability-sampling estimator used by the other two designs.

Across all nine cells, the mix increases certification in every paired query order. The mean change in residual

**Table 4:** Allocation at a 20% criterion cap, release task order. Entries after queries are marginal medians across 31 orders. Width refers to a 95% conditional hypergeometric interval for criterion microaccuracy. The mix spends up to half its cap on certification, then samples the remainder. Pure certification uses judge-first short circuit; the dash indicates that no probability-sampling accuracy interval is reported for it.

| Frame / primary | Allocation | Queries | Certified | Errors | Width (pp) |
|---|---|---|---|---|---|
| DR Gemini | SRS | 323 | 131 | 8 | 5.5 |
| | Fixed mix | 323 | 169 | 9 | 7.6 |
| | Pure certification | 323 | 275 | 1 | — |
| JB GPT-5.4, full | SRS | 4,697 | 798 | 223 | 2.2 |
| | Fixed mix | 4,697 | 1,035 | 133 | 3.0 |
| | Pure certification | 4,697 | 1,512 | 15 | — |

disagreements ranges from about $-2.3$ to $+0.8$ across the five RuVerBench cells, compared with $-91.5$ to $-58.7$ across the four JudgmentBench cells. The focal DR Gemini comparison is the only positive mean contrast. The RuVerBench changes are small in absolute task counts. Below, we test task-composition sensitivity for the focal DR Gemini comparison and both strong-judge JudgmentBench targets.

Other acquisition controls show why finding criterion errors is a poor proxy for completing verdicts. At the DR 20% cap, judge-first short circuit finds a median of 14 criterion errors and certifies 275 tasks; disagreement-first continuation finds 83 errors and certifies 160. Both leave a median of one task error. The opposing discovery and certification ranking holds in all 31 paired orders. It also occurs for the weaker JudgmentBench judge, but in none of the orders for the stronger judge. We therefore retain both primary-judge settings when assessing the value of disagreement priority.

## 4.2 Where the certificate gain goes

Let $E_0$ be the initial verdict disagreement count, and let $C_W$ and $C_R$ count certified units that were initially wrong and right. As in Proposition 1, $D$ counts repairs made before certification and $A$ counts introduced false passes.

**Proposition 3** (Certification and repair accounting)**.** *At every query endpoint,*

$$C = C_R + C_W, \qquad E^{\text{eager}} = E_0 - C_W - D + A. \tag{3}$$

Certified initially wrong units are repaired; among uncertified units, the only repairs and new errors are $D$ and $A$. For two allocations on the same frame,

$$\Delta C = \Delta C_R + \Delta C_W, \qquad \Delta E^{\text{eager}} = -\Delta C_W - \Delta D + \Delta A. \tag{4}$$

A certificate on an already correct verdict adds evidence without repairing a mistake. Conversely, eager updating can repair a verdict before the task is certified. We use this distinction to locate the source of each allocation's gain.

In DR Gemini, the mix gains 40.0 certificates on average: 40.6 on initially correct tasks and $-0.6$ on initially wrong tasks. It also loses 0.2 uncertified repairs, leaving 0.8 more residual errors. The net certification gain therefore comes from verdicts that were already right. In JudgmentBench, certification increases by 237.7 overall and by 100.5 among initially wrong annotations. The mix loses 18.9 uncertified repairs but introduces 8.5 fewer new false passes, leaving 90.1 fewer disagreements. Figure 2 displays the additive paired components.

The JudgmentBench allocation gains persist under the separate sensitivity checks. Both full and binary-only comparisons reduce error in all 399 task resamples. On the matched 206-output frame, all four acquisition/evaluation-reference combinations improve mean certification and repair at 620 queried labels. Using the same reference for acquisition and evaluation yields mean reductions of 11.7 output disagreements under A and 8.7 under B. Re-reading the other reference's query sets also improves the paired means, although a few individual orders increase disagreement. Appendix E.1 describes the reference-panel design.

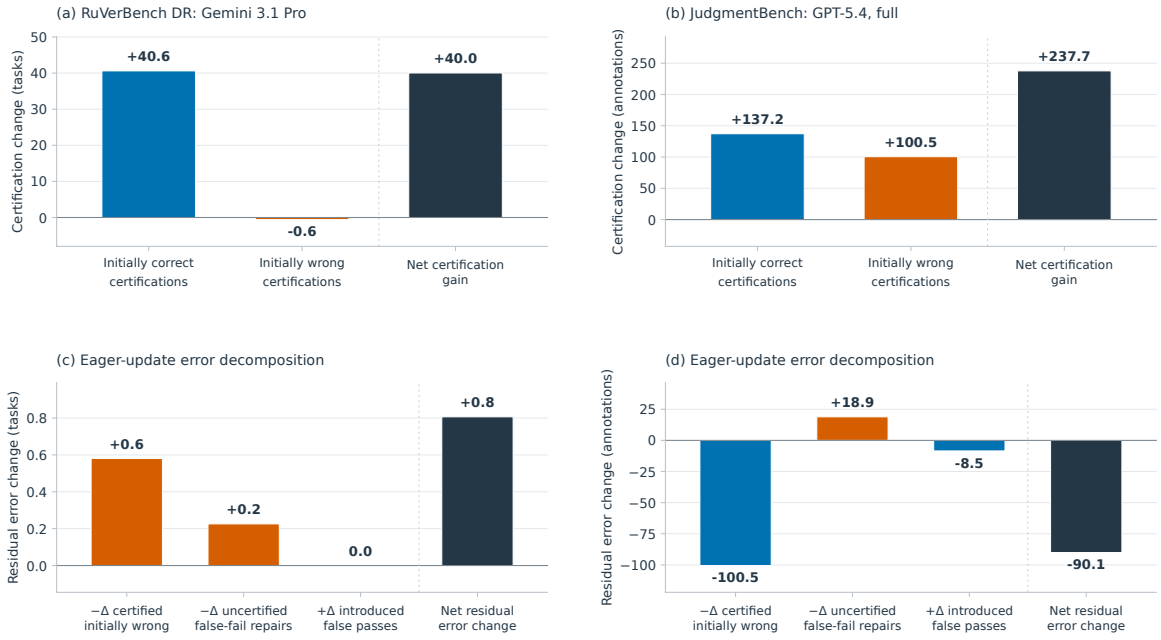



**Figure 2:** Allocation effects separated into where certificates are added and how residual disagreements change. Differences are fixed mix minus criterion SRS at a 20% budget, release task order, averaged across 31 paired orders at equal query counts. The columns show DR Gemini (284 tasks, 323 queries) and full JB GPT-5.4 (1,539 annotations, 4,697 queries). Components add before rounding. The DR error increase describes the original frame; its direction varies in the task-composition analysis.

DR gains certificates in every resample, but its repair contrast has 196 reductions, 198 increases, and five ties. We find stable certification gains in this DR Gemini comparison, while its small repair contrast changes with task composition.

## 4.3 Preserving estimation while targeting certification

The fixed mix also provides a transparent sampling estimator. If its first phase observes set $C$, the remainder contains $R = M - |C|$ bits. Draw $n$ of those bits by SRS and write $S$ for the sample. With $a_u = \mathbf{1}(P_u = Y_u)$,

$$\widehat{a} = \frac{\sum_{u \in C} a_u + (R/n) \sum_{u \in S} a_u}{M}. \tag{5}$$

Conditional on the first phase, this estimator is unbiased because the second phase is an independent probability sample. We invert hypergeometric tails for the unknown remainder agreement count and add the known first-phase total. This yields a fixed-budget conditional interval. Its justification follows from the design, in the tradition of finite-population sampling (Horvitz and Thompson, 1952), rather than from the observed coverage in 31 repeats.

Alongside SRS, we compare proportional stratification and pilot-based Neyman allocation using primary predicted PASS/FAIL strata. Pilot labels count toward the budget. Known pilot totals are added to estimated remaining stratum totals; hypergeometric count bounds are combined with Bonferroni error allocation. These controls isolate a conventional design-based alternative alongside the prediction-powered methods developed in prior work (Angelopoulos et al., 2023; Fisch et al., 2024; Kim, 2026).

At the 20% cap, proportional stratification lowers exact design RMSE by 0.03–4.9% across all nine cells: for example, AC DeepSeek changes from 1.84 to 1.77 pp. The charged-pilot Neyman design has 1.1–9.5% higher RMSE than SRS in eight cells and 0.9% lower RMSE in DR Qwen. These values integrate every possible pilot count outcome using the implemented allocation and conditional estimator. Meanwhile, the

fixed mix has an estimated 33–39% higher RMSE than SRS across the nine cells. This estimate averages exact conditional variances over 31 first-stage realizations; Appendix D.4 reports its remaining Monte Carlo error. The additional certificates therefore carry a measurable estimation cost.

Point-estimation precision and interval width also differ. In full JudgmentBench, mean guaranteed widths are 2.21 pp for SRS, 3.47 for proportional stratification, and 3.74 for pilot Neyman. Proportional stratification slightly improves RMSE while producing a wider guarantee-based interval: Bonferroni allocation and summing stratum endpoints make the construction conservative. Appendix Figure 3 shows both quantities. The original 31-order empirical RMSE remains a simulation diagnostic.

Estimating $\theta$ calls for another sampling unit. Select a fixed SRS of tasks and certify their outcomes, either by reading every rubric item or by stopping at the first failure. Both procedures observe the same sampled $Z_i$ and give the same pass-rate estimator. For 57 sampled DR tasks, median query cost falls from 319 to 67; for 308 JudgmentBench annotations, it falls from 4,690 to 1,002. Their exact design RMSEs for task pass rate are 3.54 and 1.81 pp. This fixed-unit-count design has variable label cost; it answers a different resource question from Table 4.

# 5 RQ3: What criterion summaries leave unresolved

The update mechanism depends on which failures co-occur within a task and where they lie in the query order. Criterion confusion counts do not retain these associations. We test their sufficiency by shuffling references within task-length × primary-prediction × criterion-category strata, preserving the stratum confusion counts, task lengths, predictions, and observable priorities.

For uniform criterion sampling, the full-JudgmentBench gated-minus-eager difference at a 20% cap is +12.25 annotation disagreements. This baseline uses a different acquisition policy from the disagreement-first rows in Table 2. Across 64 matched-margin configurations, its effect ranges from −9.76 to −1.48: identical criterion summaries favor the opposite update rule under equal error costs. Across nine cells, two acquisition policies (uniform criterion sampling and disagreement-first continuation), and five budgets, 44 of 90 conditions admit an opposite-sign effect. In 19 of these, all in DR, both the original and strongest reverse effect are smaller than one evaluation unit. Appendix Table 6 reports all counts and per-100-unit magnitudes.

The rearrangements test information sufficiency, whereas the resamples in Section 3 test task composition. The rearrangements change reference passes from 227 to between 108 and 132 in full JudgmentBench, jointly changing task success and error co-occurrence. Their effect ranges cannot separate those two changes or measure how often a reversal occurs in new data.

A small construction shows that changing the overall pass rate is not necessary for non-identification. Appendix B.4 gives two four-task frames with identical criterion confusion counts, predictions, rubric lengths, and reference verdicts for every task. Under uniform sampling of five of their twelve conditions, the expected gated-minus-eager differences are +7/66 and −7/132. The frames differ only in the location of a reference label within the same prediction stratum. This is a logical counterexample; we have not performed pass-total-constrained rearrangements of the empirical JudgmentBench frame.

We next hold the reference frame fixed and vary what information is disclosed. In the query-only model, let $n_P$ and $n_F$ denote certified PASS and FAIL counts. Unrestricted completions give the sharp endpoint range

$$\theta \in \left[\frac{n_P}{N}, 1 - \frac{n_F}{N}\right]. \tag{6}$$

Each unresolved task can independently be completed to PASS or FAIL, so the endpoints are attainable. More certificates narrow this range. A probability sample can nevertheless give a useful statistical estimate of $\theta$ while many individual tasks remain unresolved: logical identification and sampling precision answer different questions (Rao and Callison-Burch, 2026; Chen et al., 2026).

Additional trusted summaries change the feasible set. Our disclosure replay begins with reference-containing confusion margins grouped by task length and prediction, refines them by criterion category, and then restores labels linked to whole tasks. Binary mixed-integer optimization computes attainable minimum and maximum pass totals. For DR Gemini, the initial margins leave the pass rate between 4.23% and 16.90%, a width of 12.68 pp. These margins comprise 38 disclosed count cells, not 38 new reference queries. Full link restoration

recovers the observed pass rate. A trusted reference task-PASS total $T$ would instead identify the aggregate directly as $\theta = T/N$. The initial disclosure does not include $T$; our ladder compares information sets rather than establishes a minimal release. Knowing the aggregate alone generally leaves individual certificates and update-policy effects unresolved.

Appendix F gives a complete two-task example of how a trusted total and linked observations jointly identify the pass rate. Aggregate identification can still leave individual verdicts unresolved.

The practical implication is to report the information that supports the intended conclusion. Linked criterion records support analysis of update and certification behavior; criterion-level confusion summaries alone generally do not. RuVerBench already releases linked records, which make this replay possible. Preserving these links in downstream validation reports lets evaluators examine consequences that aggregate agreement conceals.

# 6 Related work

Multi-predicate screening studies condition-level crowd queries, posterior stopping, and downstream classification loss (Krivosheev et al., 2018a,b, 2020). Sequential system testing and stochastic Boolean-function evaluation formalize how costly component checks determine an outcome (Ünlüyurt, 2004; Gkenosis et al., 2022; Ghuge et al., 2026). Structured agent evaluation also uses node-level checks and short circuit (Gou et al., 2025). Our logical certificates use these foundations. The empirical extension is to compare their role in repairing an already available judge verdict on matched queried sets.

Selective correction and local error propagation have close precedents. ActiveClean separates sample selection from model updating and shows why mixed clean and dirty data can distort learning (Krishnan et al., 2016). Query-Guided Verification traces AND/OR query provenance, including cases where lowering a component's error probability raises a worst-case output score (Schreiber and Amsterdamer, 2026). Its maximum-error score uses independent verification errors and differs from our realized FP/FF counts after reference-bit replacement. Correction-and-corruption analysis uses paired outcomes, exact accounting, and estimated rates to study selective application (Reitich, 2026). Our comparison applies these correction principles to the timing of rubric-verdict updates: identical queried sets isolate waiting for certification, and budget curves locate when prevented errors or delayed repairs dominate. Reference and task-composition analyses then test the stability of those empirical effects.

Failure discovery and performance estimation have long motivated different sample choices. DeepEST combines failure-oriented selection with population estimation, while recent multi-objective test selection compares discovery, estimation, and retraining (Guerriero et al., 2021; Zhang et al., 2026). Active Testing and prediction-assisted inference use partial trusted labels to estimate aggregate performance (Nguyen et al., 2018; Chaganty et al., 2018; Angelopoulos et al., 2023). StratPPI studies stratified allocation, and Kim uses a two-phase doubly robust design and pilot information to plan human-review requirements (Fisch et al., 2024; Kim, 2026). We add the joint accounting of certification and verdict repair, while using conventional sampling controls for estimation.

Different guarantees govern different decisions. Trust or Escalate calibrates acceptance and escalation of pairwise judgments to control disagreement risk among accepted judgments under its calibration assumptions (Jung et al., 2025); audit-efficient best-agent identification targets system selection (Ao et al., 2026). A logical certificate instead resolves a particular conjunction from observed reference bits. Evaluation-protocol and partial-identification work further distinguishes the conclusions licensed by aggregated information (Rao and Callison-Burch, 2026; Chen et al., 2026).

Finally, human uncertainty affects automatic-evaluator assessment (Elangovan et al., 2025). Physician disagreement in HealthBench has been studied through variance decomposition and prediction (Borgohain and Mariathas, 2026). Our reference sensitivity carries this concern through the checking protocol: it fixes both judges, changes the reference panel, and separates re-reading a fixed query set from rerunning acquisition. The resulting differences concern protocol conclusions as well as initial agreement.

# 7 Discussion and limitations

The results connect each evaluation objective to a different use of reference information. For criterion accuracy, probability sampling supports estimation. Proportional strata give small RMSE improvements in our comparison, but their guaranteed intervals are wider because our implementation combines separate stratum bounds conservatively. This is a cost of the chosen interval construction, not an inherent cost of stratification. For individual certificates and logical pass-rate bounds, concentrating queries within tasks resolves outcomes quickly. For repair, the relevant interval runs from clearing a mistaken rejection to obtaining a complete certificate. Serial task completion limits the eager–gated difference to one verdict; dispersed checks leave room for a much larger backlog.

For false-fail repair, the relevant coverage is whether flags reach every erroneous rejection. In full JB GPT-5.4, all wrong rejections are flagged in just two of 220 initial false fails, compared with 36 of 67 under binary-only scoring. Item-level coverage is 7.7% for erroneous penalty rejections and 73.8% for erroneous binary rejections. Consequently, 218 full-target repairs require random continuation. Low rejection coverage explains why these repairs occur late; it does not itself explain a larger eager–gated delay. For a fixed remaining sampling design, earlier removal of the last erroneous rejection opens the delay window sooner, while checking the remaining passing conditions closes it. Appendix B.3 reports the full coverage partition and reference diagnostics.

For false-pass and false-fail costs $c_{FP}$ and $c_{FF}$, the expected gated-minus-eager loss is $c_{FF}\mathbb{E}[D] - c_{FP}\mathbb{E}[A]$. When $c_{FF} > 0$ and $\mathbb{E}[A] > 0$, gating has lower expected loss exactly when $c_{FP}/c_{FF} > \mathbb{E}[D]/\mathbb{E}[A]$. Using the unrounded expectations underlying Table 2, the break-even ratio is approximately 0.725 at a 20% budget and 9.1 at 84%. At the 84% budget in this comparison, waiting for certificates pays off only when a false pass costs more than roughly nine false fails. These thresholds describe the saved frame. Prospective selection would require affordable estimates of the relevant error structures, a charged pilot budget, and confirmation on unused data.

The study has four limits. First, all repairs and certificates refer to released labels. We do not adjudicate the raters, and a rubric certificate only establishes the retained conditions. Second, the all-pass targets are constructed from two data families. Coding results broaden the observed error regimes, but do not establish transfer to a native all-pass benchmark. Third, the experiments are retrospective. Repeated query orders do not create new tasks, and resampling 30 curated legal base tasks measures composition sensitivity without making them representative of every future environment. Reference and composition checks are separate, rather than a joint uncertainty interval. Fourth, budgets count revealed labels. They omit expert time, auxiliary-judge costs, and the work of resolving reference disagreement.

The complete-reference calculations are explanations of the observed frames. A new environment may have different task lengths, pass rates, error overlap, or query priorities. A useful extension would test a prespecified prediction of protocol performance on unused tasks or on a native all-pass benchmark. Such a study would address transfer of the mechanism; a claim about expert labor would additionally need measured annotation costs. The present paper supplies the decomposition and reproducible comparisons with which to formulate those tests.

A system could instead publish PASS, FAIL, or UNRESOLVED and withhold every uncertified verdict. The certificate counts already describe the coverage of that reporting rule. Our two update rules answer a different operational question: how to maintain an existing binary verdict during checking. Comparing that practice with abstention requires a cost for UNRESOLVED, alongside false-pass and false-fail costs. Hybrid update rules would likewise need a specified objective and an independently evaluated selection rule.

# 8 Conclusion

A reference check can improve an estimate, establish an outcome, or repair a verdict, and these gains have different information requirements. Matched-query comparisons show that the cost of waiting for a certificate depends on the interval between correcting erroneous rejections and completing the task. Allocation comparisons show that additional certificates improve repair only through the tasks and error events they resolve. Reference and task-composition checks distinguish persistent effects from small, unstable contrasts. Reference labels should buy the evidence the evaluation needs: probability samples for population estimates, completed

checks for individual certificates, and timely correction for verdict repair. Choosing among these objectives also means choosing where to query and when to update.

## Data, code, and AI assistance

The public project is hosted at https://github.com/Itsxuchen/RefEval. This local review draft corresponds to package version `0.3.0-rc.3`; publication of its matching snapshot awaits author approval. Reproduction requires no model API calls. The package preserves upstream attribution and data terms and redistributes compact numerical records. JudgmentBench judge predictions are source-published; RuVerBench predictions come from our saved runs.

Codex and Claude assisted with code, analysis checks, literature searches, and drafting. We checked generated material against source papers, code, and saved data. The named human author is responsible for the manuscript.

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# A Finite reference frames and verdict accounting

Let $\mathcal{J}_i$ be the nonempty set of criterion slots for evaluation unit $i$, with $k_i = |\mathcal{J}_i|$, $N$ units, and $M = \sum_i k_i$ slots. Slots belong to exactly one unit. The released reference and primary prediction for slot $u$ are $Y_u, P_u \in \{0, 1\}$, where one denotes PASS. Throughout this appendix they are fixed finite-population values. Define

$$Z_i = \prod_{u \in \mathcal{J}_i} Y_u, \qquad \widehat{Z}_i^0 = \prod_{u \in \mathcal{J}_i} P_u, \qquad \theta = \frac{1}{N} \sum_i Z_i. \tag{7}$$

Criterion microagreement is the distinct target $a = M^{-1} \sum_u \mathbf{1}\{Y_u = P_u\}$. An error in the following equations means disagreement with this particular reference. No equation identifies which of two disagreeing human raters is substantively correct.

For JudgmentBench full strict, a positive binary item passes when its original value is one, and a negative-weight occurrence-count item passes when its original count is zero. The loader checks that their conjunction equals attaining the rubric's maximum score. Binary-only retains positive binary items and removes occurrence-count items. This changes both the success event and the criterion population. The original unit is an output-by-rater annotation; the alternative-reference analysis instead weights distinct outputs equally. RuVerBench units use the criteria retained in its release.

## A.1 Certificates and update rules

Let $Q \subseteq \bigcup_i \mathcal{J}_i$ be the queried slots and $Q_i = Q \cap \mathcal{J}_i$. In the query-only information model, a FAIL certificate is an observed reference failure, and a PASS certificate requires observing every slot as PASS. Write

$$C_i^- = \mathbf{1}\{\exists u \in Q_i : Y_u = 0\}, \tag{8}$$

$$C_i^+ = \mathbf{1}\{Q_i = \mathcal{J}_i,\ Z_i = 1\}, \qquad C_i = C_i^- + C_i^+. \tag{9}$$

Eager updating substitutes every queried reference value immediately:

$$\widehat{Z}_i^E(Q) = \prod_{u \in \mathcal{J}_i} \big(\mathbf{1}\{u \in Q\} Y_u + \mathbf{1}\{u \notin Q\} P_u\big). \tag{10}$$

Certificate-gated updating uses $Z_i$ once certified and otherwise retains $\widehat{Z}_i^0$:

$$\widehat{Z}_i^G(Q) = C_i Z_i + (1 - C_i)\widehat{Z}_i^0. \tag{11}$$

Writing $Z_i$ here is shorthand for the value logically determined by a certificate, not access to an unqueried reference. The implementation obtains that value from the queried bits. Gating can retain an uncertified initial PASS; it is not a protocol that publishes only certified outcomes.

## A.2 Acquisition implementations

The seven acquisition rules use a common slot map. Uniform criterion sampling follows a global random priority. Three serial short-circuit rules visit tasks in the stated order and visit their criteria in natural slot order, random order, or predicted-FAIL-first order, respectively. They stop the current task at its first observed reference failure or after completing its rubric. Disagreement-only visits flagged criteria within each task and stops after the flagged pool is exhausted; it can leave budget unused. Disagreement-then-random uses that same flagged prefix, followed by a global random ordering of all unflagged criteria. Disagreement-prioritized short circuit visits, within a task, flagged slots first, then unflagged predicted FAIL slots, then the remaining predicted PASS slots; its stopping rule is the same serial certification rule.

For each DR primary, the auxiliaries are the other two saved DR judges among Gemini 3.1 Pro, Qwen-plus and GPT-5.4 low. Each AC primary has the other saved AC judge as its auxiliary: Qwen-plus or DeepSeek v4-pro. JudgmentBench pairs GPT-5.4 with GPT-5.4-mini within the same scoring target. The flag is the union of disagreements with available auxiliaries. Their predictions are treated as available signals; their acquisition cost is not included in the reference-bit budget.

The three task orders are release-relative order, ascending primary predicted-failure count, and a random task order. Seeds 1–31 randomize ties within the declared priorities. A separate canonical seed-zero realization uses natural slot order for priority ties while retaining random priorities for explicitly random policies and the random task order. Purpose-specific streams derive from root seed 20260913, the replicate seed and a SHA256 hash of the stream's name. Release-relative order refers to the loader's retained unit ordering, not a randomized sample of deployment tasks. The analytical mechanism calculations below hold this order fixed.

### A.3 Exact event decomposition

Define predicted-failure slots $F_i = \{u \in \mathcal{J}_i : P_u = 0\}$ and reference-failure slots $T_i = \{u \in \mathcal{J}_i : Y_u = 0\}$. An initially correct FAIL is susceptible to a new false PASS only if $F_i$ and $T_i$ are nonempty and disjoint. Its new-error event is

$$A_i(Q) = \mathbf{1}\{F_i \neq \varnothing,\ T_i \neq \varnothing, F_i \cap T_i = \varnothing,\ F_i \subseteq Q, T_i \cap Q = \varnothing\}. \tag{12}$$

An initially false FAIL has $T_i = \varnothing$ and $F_i \neq \varnothing$. Eager updating repairs it once all its predicted failures have been checked. Gating delays that repair until every criterion has been checked. The delayed-repair event is therefore

$$D_i(Q) = \mathbf{1}\{T_i = \varnothing,\ F_i \neq \varnothing, F_i \subseteq Q,\ \mathcal{J}_i \not\subseteq Q\}. \tag{13}$$

Let $A(Q) = \sum_i A_i(Q)$, $D(Q) = \sum_i D_i(Q)$, and $E_r(Q) = \sum_i \mathbf{1}\{\widehat{Z}_i^r(Q) \neq Z_i\}$ for $r \in \{E, G\}$. Then

$$E_G(Q) - E_E(Q) = D(Q) - A(Q). \tag{14}$$

**Derivation.** Initially correct PASS units have every prediction and reference equal to one, so either update remains correct. An initially false PASS is repaired exactly when a reference failure is queried, which also certifies FAIL; both updates therefore agree. For an initially correct FAIL, an unqueried predicted failure or a queried reference failure prevents eager PASS. Eager can introduce a false PASS precisely in (12), while gating retains the correct FAIL. For an initially false FAIL, the two rules disagree precisely in (13). Summing these four exhaustive initial classes proves (14) and Proposition 1. In particular, correct reference substitutions cannot introduce a false FAIL under conjunction.

If false PASS and false FAIL have fixed nonnegative costs $c_{\mathrm{FP}}$ and $c_{\mathrm{FF}}$, their endpoint loss difference is $c_{\mathrm{FF}}D - c_{\mathrm{FP}}A$. Thus equal-weight error comparisons are one particular loss choice. Along a growing query transcript, gated error cannot increase: every changed verdict is certified relative to the fixed reference. Its difference from eager error can increase because eager may repair false FAILs earlier. At zero queries and at a census the difference is zero.

Both rules agree on untouched and certified units. If $U(Q)$ units have been partially queried but remain uncertified, then $|E_G(Q) - E_E(Q)| \leq U(Q)$. A serial short-circuit policy that completes or certifies a task before starting the next has $U(Q) \leq 1$ at every prefix. This explains why changing the update rule has little room to change the endpoint under that acquisition structure, even when dispersed querying has a large effect.

## B Finite-budget expectations over query order

For uniform sampling of $h$ slots without replacement from a group of size $G$, the probability of including $r$ specified slots and excluding $f$ other specified slots is

$$H(G, h; r, f) = \frac{\binom{G-r-f}{h-r}}{\binom{G}{h}}. \tag{15}$$

The specified inclusion and exclusion sets are disjoint. We set the probability to zero when $r + f > G$, $r > h$, or $f > G - h$; in the empty case $H(0, 0; 0, 0) = 1$. Equation (15) follows by choosing the other $h - r$ observed slots from the $G - r - f$ unrestricted slots. It is a sampling calculation and assumes no independence of reference errors.

For a uniform $m$-of-$M$ criterion query set, the two expectations are

$$\mathbb{E}[A(Q)] = \sum_{i:\, F_i, T_i \neq \varnothing,\, F_i \cap T_i = \varnothing} H(M, m; |F_i|, |T_i|), \tag{16}$$

$$\mathbb{E}[D(Q)] = \sum_{i:\, T_i = \varnothing,\, F_i \neq \varnothing} \{H(M, m; |F_i|, 0) - H(M, m; k_i, 0)\}. \tag{17}$$

The complete-rubric event is contained in the event of observing every predicted failure, which justifies the subtraction in the second line. Linearity of expectation gives the expected update difference even though events for different tasks are dependent under a fixed global budget.

## B.1 Prioritized groups

The same calculation applies to a predetermined sequence of disjoint query groups $\mathcal{G}_1, \ldots, \mathcal{G}_L$. Every slot in an earlier group is queried before any slot in a later group, and order within each group is uniformly random. At an interior cap $m$, let $\ell$ be the group satisfying

$$\sum_{g<\ell} |\mathcal{G}_g| \leq m < \sum_{g\leq\ell} |\mathcal{G}_g|, \qquad h = m - \sum_{g<\ell} |\mathcal{G}_g|.$$

For a required set $R$ and forbidden set $S$, the event $R \subseteq Q$, $S \cap Q = \varnothing$ has probability zero if a required slot lies after $\ell$, a forbidden slot lies before $\ell$, or the two sets overlap. Otherwise its probability is

$$H\big(|\mathcal{G}_\ell|, h; |R \cap \mathcal{G}_\ell|, |S \cap \mathcal{G}_\ell|\big). \tag{18}$$

At a complete group boundary, the next group has $h = 0$. At a census the event is determined directly by the complete query set. Substituting this event probability for each term in (17) gives the grouped-order expectations.

For disagreement-then-random, a criterion is flagged when any available auxiliary judge disagrees with the primary prediction. Each task's flagged criteria form one group, tasks follow release order, and the final group contains all unflagged criteria. The flagged tier is therefore not a single global random sample. These groups depend on saved predictions and task order, not hidden references. The analytical grid covers uniform criterion and disagreement-then-random querying at 101 shares from zero to one. It does not apply the predetermined-group formula to adaptive short-circuit paths. Budget caps use the implementation's nearest-integer rule $m = \text{round}(bM)$, with ties to an even integer.

All expectations condition on the full reference, predictions and priority groups. They explain the observed frames rather than provide a free prospective policy selector. The 31 random-order means provide a Monte Carlo check against these expectations; their standard error is the sample standard deviation divided by $\sqrt{31}$, not uncertainty from sampling new tasks.

## B.2 Task-order comparisons

Table 5 reports the three task orders for disagreement-first continuation. The five-percent budget often ends inside the flagged tier, so changing which task is visited first changes the endpoint effect. When the cap covers every flag, the remaining population is independent of the order of those completed queries. Its uniform continuation then has the same distribution. Purpose-specific random streams also make the saved paired continuations identical across task orders after this boundary.

For full JB GPT-5.4, all 4,316 flags are covered at 18.4% of the 23,487-criterion frame. At 84%, each of the three task orders has a saved 31-realization mean of 97.65 disagreements; the common analytical expectation is 96.33. Their difference is query-randomization Monte Carlo error. Uniform criterion sampling has no task-priority stage. For serial short-circuit rules, the unresolved-task bound in Proposition 2 applies to every task order.

**Table 5:** Mean gated-minus-eager disagreements over seeds 1–31 for disagreement-first continuation. Columns distinguish release order (R), ascending predicted-failure count (F), and random task order (X). Counts use tasks in DR/AC and annotations in JB. A zero at an early budget can mean that no affected task has yet been reached. These are query-order means, not task-composition uncertainty intervals.

| | 5% budget | | | 20% budget | | |
|---|---|---|---|---|---|---|
| Frame / judge | R | F | X | R | F | X |
| DR Gemini | 0.00 | 2.00 | 0.48 | 2.00 | 2.00 | 2.00 |
| DR Qwen | 0.00 | 2.10 | 0.71 | 3.00 | 3.00 | 3.00 |
| DR GPT-5.4 | 0.00 | 2.00 | 0.61 | 2.00 | 2.00 | 2.00 |
| AC Qwen | 2.00 | 0.55 | 2.45 | 5.52 | 5.52 | 5.52 |
| AC DeepSeek | 6.00 | 8.90 | 5.03 | 11.23 | 11.23 | 11.23 |
| JB GPT-5.4 full | 0.00 | -1.00 | -0.97 | -2.06 | -2.06 | -2.06 |
| JB mini full | -1.00 | -13.00 | -3.06 | -11.68 | -11.68 | -11.68 |
| JB GPT-5.4 binary | 4.00 | 4.00 | 1.65 | 8.29 | 8.29 | 8.29 |
| JB mini binary | 1.00 | 0.00 | -0.81 | -0.32 | -0.32 | -0.32 |

## B.3 Peer evidence and disagreement-flag coverage

Table 3 partitions initial false fails by other human raters of the same output, retaining the original annotation-specific judge vectors. A peer must have a different rater ID. The categories are at least one peer FAIL, all available peers PASS, and no peer. No annotation is removed and no query budget, weight, or priority is changed.

At the 84% cap, all 4,316 flagged criteria have been read and the continuation selects $h = 15{,}413$ of the remaining $G = 19{,}171$ criteria uniformly. For initial false fail $i$, let $r_i$ count unflagged predicted failures and let $k_i^u$ count all its unflagged criteria. Its expected contribution is

$$d_i = \frac{\binom{G-r_i}{h-r_i}}{\binom{G}{h}} - \frac{\binom{G-k_i^u}{h-k_i^u}}{\binom{G}{h}}. \tag{19}$$

The first term is eager repair; the second is completed PASS certification. Summing $d_i$ within the three peer groups gives 15.4554, 8.6997, and 84.0610, respectively, totaling 108.2161. This decomposes delayed repairs $D$, not the net update contrast $D - A$.

The same fixed-query calculation can partition delay by the original rejected item types. The 156 false fails with only penalty rejections contribute 77.8711 expected delays; the 64 with both binary and penalty rejections contribute 30.3450. These are descriptive contributions on the original frame. Removing penalty items changes the scoring target and query design, so this partition is not the effect of deleting those items.

The two terms in Equation (19) must be interpreted together. Holding $G, h, k_i^u$ fixed, decreasing the number $r_i$ of remaining wrong rejections increases the probability of eager repair, hence increases delay relative to certification. Changing the flagged set can also change $G, h, k_i^u$. Even when every wrong rejection is already queried, delay is $1 - \Pr(\text{remaining rubric complete})$, rather than necessarily close to one. Under the displayed continuation design this is 0.1960 for one remaining item and 0.8872 for ten. The timing of both repair and certification determines the endpoint comparison.

Coverage is measured after every disagreement flag has been queried. Among 220 initial false fails, 79 have no flagged item, 80 have flags only on predicted-PASS items, 59 have some but not all erroneous rejections flagged, and two have all erroneous rejections flagged. None has its whole rubric flagged. The false-fail repair condition requires every predicted failure to be queried, whereas a PASS certificate requires every criterion. At the item level, 163 of 221 erroneous binary rejections and 21 of 274 erroneous penalty rejections are flagged. The corresponding all-rejections-covered count for the changed binary-only target is 36 of 67 initial false fails.

We also made an exploratory item-level check within the repeated-rating subset of these original full-target false fails. Both judges reject 62 penalty instances that the current reference passes; another rater rejects 15 of them. The analogous counts for binary instances are six of 11. Among flagged erroneous rejections, the counts are zero of three penalty instances and 20 of 32 binary instances. An instance is a criterion within an original annotation; repeated annotations share outputs and criteria. The sparse, selected groups do not

establish that shared penalty rejections are the most unreliable references, and peer disagreement does not establish which rating is correct. This check changes neither references nor policies.

## B.4 A counterexample with fixed reference verdicts

The following constructed frames isolate the information question from a change in aggregate pass rate. Each has four tasks of three binary criteria, one criterion category, and the same primary predictions. Only the first reference bit of tasks 1 and 4 is exchanged:

| Task | Prediction $P_i$ | Reference $Y_i^{(I)}$ | Reference $Y_i^{(II)}$ |
|---|---|---|---|
| 1 | $(0,1,1)$ | $(0,0,0)$ | $(1,0,0)$ |
| 2 | $(0,0,1)$ | $(1,1,1)$ | $(1,1,1)$ |
| 3 | $(1,1,1)$ | $(0,0,0)$ | $(0,0,0)$ |
| 4 | $(0,0,0)$ | $(1,0,0)$ | $(0,0,0)$ |

Both frames have criterion-level $(TP, FP, FN, TN) = (1, 5, 3, 3)$. All tasks have equal length and category, so the exchange also preserves the stratified criterion confusion counts used in the empirical test. Both reference verdict vectors are $(0, 1, 0, 0)$; even task-level confusion counts and the PASS total $T = 1$ agree.

Apply the query-only update rules to a uniform five-of-twelve sample. Task 2 is the only initial false fail in either frame. With $H$ as defined in Equation (15),

$$\mathbb{E}[D] = H(12, 5; 2, 0) - H(12, 5; 3, 0) = \frac{7}{66}.$$

In frame I, every initially correct rejection retains a correctly predicted reference failure, so $\mathbb{E}[A] = 0$. In frame II, task 1 can become a false pass when its first criterion is queried and its other two are not:

$$\mathbb{E}[A] = H(12, 5; 1, 2) = \frac{7}{44}.$$

Thus the expected gated-minus-eager effects are $7/66$ and $-7/132$. The implementation check enumerates all $\binom{12}{5} = 792$ queried sets and directly recomputes both verdicts. The identical task truths are a constraint on this construction, not additional certificates handed to the query-only controllers. This example establishes non-identification of their effects even at a fixed task PASS total; it does not measure the range of such effects in the JudgmentBench data.

## B.5 Matched-margin alternatives

We construct 64 alternative assignments in each of nine analysis cells by permuting reference labels within strata defined by task length, primary prediction and criterion category or scoring mode. This preserves every stratum's prediction/reference confusion counts, the task slots and lengths, and all primary and auxiliary predictions. Root seed 20261007 and a dataset-specific hash define the random streams. Both analytical policies are evaluated at shares $0.05, 0.20, 0.50, 0.84, 0.95$, including originally observed focal and peak regions. An alternative assignment with a different update effect demonstrates that the retained summaries do not determine that effect. These are retrospective counterexamples, not a null test for random reference noise. The permutations can also change reference task pass counts, initial task errors and their joint configuration, so they do not isolate a causal effect of error correlation.

Table 6 gives the entire saved grid. Each range is the minimum and maximum over 64 configurations, not an identification bound over all compatible assignments or a confidence interval. The 19 opposite-sign conditions with both original and strongest reversed effects below one unit are all DR cells. This threshold describes magnitude, not statistical significance or a prespecified practical cutoff. Counts retain each cell's original evaluation unit: tasks for DR/AC and output-by-rater annotations for JB. The per-100 columns facilitate scale comparisons across frames.

**Table 6:** Update-effect magnitudes for all 90 matched-margin conditions. Positive $\Delta = \mathbb{E}[D - A]$ favors eager updating under equal error costs. U: uniform criterion sampling; D: disagreement-first continuation. Cap is percent of criteria; Opp. counts configurations with a nonzero sign opposite to the original, out of 64. Full and binary denote scoring targets.

| | | | Disagreements | | Per 100 units | | |
|---|---|---|---|---|---|---|---|
| Cell | Policy | Cap | Original | Configuration range | Original | Configuration range | Opp. |
| AC DeepSeek | D | 5 | 6.00 | [-1.00, 6.00] | 2.86 | [-0.48, 2.86] | 2 |
| AC DeepSeek | D | 20 | 11.15 | [2.07, 12.27] | 5.31 | [0.99, 5.84] | 0 |
| AC DeepSeek | D | 50 | 10.48 | [3.35, 12.21] | 4.99 | [1.59, 5.81] | 0 |
| AC DeepSeek | D | 84 | 5.29 | [2.07, 6.93] | 2.52 | [0.98, 3.30] | 0 |
| AC DeepSeek | D | 95 | 1.90 | [0.78, 2.62] | 0.90 | [0.37, 1.25] | 0 |
| AC DeepSeek | U | 5 | 0.60 | [0.21, 0.89] | 0.29 | [0.10, 0.42] | 0 |
| AC DeepSeek | U | 20 | 2.40 | [0.97, 3.47] | 1.14 | [0.46, 1.65] | 0 |
| AC DeepSeek | U | 50 | 5.24 | [2.35, 7.32] | 2.50 | [1.12, 3.48] | 0 |
| AC DeepSeek | U | 84 | 4.16 | [1.78, 5.76] | 1.98 | [0.85, 2.74] | 0 |
| AC DeepSeek | U | 95 | 1.66 | [0.68, 2.32] | 0.79 | [0.33, 1.10] | 0 |
| AC Qwen | D | 5 | 2.00 | [-2.00, 5.00] | 0.95 | [-0.95, 2.38] | 6 |
| AC Qwen | D | 20 | 5.37 | [-1.21, 5.77] | 2.56 | [-0.58, 2.75] | 8 |
| AC Qwen | D | 50 | 6.01 | [1.22, 7.56] | 2.86 | [0.58, 3.60] | 0 |
| AC Qwen | D | 84 | 3.65 | [1.35, 5.64] | 1.74 | [0.64, 2.68] | 0 |
| AC Qwen | D | 95 | 1.39 | [0.55, 2.26] | 0.66 | [0.26, 1.08] | 0 |
| AC Qwen | U | 5 | 0.35 | [0.06, 0.75] | 0.17 | [0.03, 0.36] | 0 |
| AC Qwen | U | 20 | 1.42 | [0.41, 2.97] | 0.68 | [0.19, 1.41] | 0 |
| AC Qwen | U | 50 | 3.26 | [1.44, 6.60] | 1.55 | [0.69, 3.14] | 0 |
| AC Qwen | U | 84 | 2.86 | [1.59, 5.53] | 1.36 | [0.76, 2.63] | 0 |
| AC Qwen | U | 95 | 1.19 | [0.69, 2.26] | 0.57 | [0.33, 1.08] | 0 |
| DR GPT-5.4 | D | 5 | 0.00 | [-1.00, 1.00] | 0.00 | [-0.35, 0.35] | 0 |
| DR GPT-5.4 | D | 20 | 2.00 | [-1.00, 2.00] | 0.70 | [-0.35, 0.70] | 8 |
| DR GPT-5.4 | D | 50 | 1.84 | [-0.62, 1.93] | 0.65 | [-0.22, 0.68] | 8 |
| DR GPT-5.4 | D | 84 | 0.95 | [-0.32, 1.27] | 0.33 | [-0.11, 0.45] | 7 |
| DR GPT-5.4 | D | 95 | 0.35 | [-0.12, 0.51] | 0.12 | [-0.04, 0.18] | 7 |
| DR GPT-5.4 | U | 5 | 0.10 | [-0.10, 0.15] | 0.04 | [-0.03, 0.05] | 11 |
| DR GPT-5.4 | U | 20 | 0.39 | [-0.32, 0.59] | 0.14 | [-0.11, 0.21] | 11 |
| DR GPT-5.4 | U | 50 | 0.84 | [-0.50, 1.28] | 0.30 | [-0.18, 0.45] | 10 |
| DR GPT-5.4 | U | 84 | 0.67 | [-0.27, 1.01] | 0.24 | [-0.09, 0.36] | 10 |
| DR GPT-5.4 | U | 95 | 0.27 | [-0.10, 0.41] | 0.10 | [-0.03, 0.14] | 10 |
| DR Gemini | D | 5 | 0.00 | [-1.00, 1.00] | 0.00 | [-0.35, 0.35] | 0 |
| DR Gemini | D | 20 | 2.00 | [-2.00, 2.00] | 0.70 | [-0.70, 0.70] | 10 |
| DR Gemini | D | 50 | 1.62 | [-1.01, 1.85] | 0.57 | [-0.36, 0.65] | 10 |
| DR Gemini | D | 84 | 0.87 | [-0.36, 1.03] | 0.31 | [-0.13, 0.36] | 9 |
| DR Gemini | D | 95 | 0.34 | [-0.12, 0.40] | 0.12 | [-0.04, 0.14] | 9 |
| DR Gemini | U | 5 | 0.10 | [-0.10, 0.15] | 0.03 | [-0.03, 0.05] | 11 |
| DR Gemini | U | 20 | 0.36 | [-0.32, 0.54] | 0.13 | [-0.11, 0.19] | 11 |
| DR Gemini | U | 50 | 0.73 | [-0.50, 1.00] | 0.26 | [-0.18, 0.35] | 10 |
| DR Gemini | U | 84 | 0.62 | [-0.27, 0.89] | 0.22 | [-0.09, 0.31] | 8 |
| DR Gemini | U | 95 | 0.26 | [-0.10, 0.39] | 0.09 | [-0.03, 0.14] | 8 |

**Table 6:** Update-effect magnitudes (continued). Definitions and units as in the first panel.

| | | | Disagreements | | Per 100 units | | |
|---|---|---|---|---|---|---|---|
| Cell | Policy | Cap | Original | Configuration range | Original | Configuration range | Opp. |
| DR Qwen | D | 5 | 0.00 | [0.00, 0.00] | 0.00 | [0.00, 0.00] | 0 |
| DR Qwen | D | 20 | 3.00 | [-3.00, 3.00] | 1.06 | [-1.06, 1.06] | 33 |
| DR Qwen | D | 50 | 2.11 | [-1.64, 2.42] | 0.74 | [-0.58, 0.85] | 32 |
| DR Qwen | D | 84 | 0.76 | [-0.48, 1.32] | 0.27 | [-0.17, 0.46] | 28 |
| DR Qwen | D | 95 | 0.25 | [-0.18, 0.50] | 0.09 | [-0.06, 0.18] | 28 |
| DR Qwen | U | 5 | -0.05 | [-0.29, 0.21] | -0.02 | [-0.10, 0.07] | 23 |
| DR Qwen | U | 20 | -0.16 | [-0.96, 0.86] | -0.06 | [-0.34, 0.30] | 23 |
| DR Qwen | U | 50 | -0.25 | [-1.50, 1.81] | -0.09 | [-0.53, 0.64] | 32 |
| DR Qwen | U | 84 | -0.13 | [-0.81, 1.23] | -0.05 | [-0.28, 0.43] | 37 |
| DR Qwen | U | 95 | -0.05 | [-0.29, 0.49] | -0.02 | [-0.10, 0.17] | 38 |
| JB GPT-5.4 full | D | 5 | 0.00 | [-2.00, 1.00] | 0.00 | [-0.13, 0.06] | 0 |
| JB GPT-5.4 full | D | 20 | -1.97 | [-7.32, -1.57] | -0.13 | [-0.48, -0.10] | 0 |
| JB GPT-5.4 full | D | 50 | 44.75 | [-2.64, 13.40] | 2.91 | [-0.17, 0.87] | 1 |
| JB GPT-5.4 full | D | 84 | 96.33 | [26.77, 41.56] | 6.26 | [1.74, 2.70] | 0 |
| JB GPT-5.4 full | D | 95 | 61.07 | [15.02, 23.69] | 3.97 | [0.98, 1.54] | 0 |
| JB GPT-5.4 full | U | 5 | 1.26 | [-4.81, -2.52] | 0.08 | [-0.31, -0.16] | 64 |
| JB GPT-5.4 full | U | 20 | 12.25 | [-9.76, -1.48] | 0.80 | [-0.63, -0.10] | 64 |
| JB GPT-5.4 full | U | 50 | 53.12 | [7.27, 22.58] | 3.45 | [0.47, 1.47] | 0 |
| JB GPT-5.4 full | U | 84 | 89.31 | [28.37, 41.99] | 5.80 | [1.84, 2.73] | 0 |
| JB GPT-5.4 full | U | 95 | 54.22 | [14.83, 22.42] | 3.52 | [0.96, 1.46] | 0 |
| JB GPT-5.4 binary | D | 5 | 4.00 | [-15.00, 0.00] | 0.26 | [-0.97, 0.00] | 62 |
| JB GPT-5.4 binary | D | 20 | 8.22 | [-23.84, -2.72] | 0.53 | [-1.55, -0.18] | 64 |
| JB GPT-5.4 binary | D | 50 | 28.94 | [-2.88, 15.58] | 1.88 | [-0.19, 1.01] | 6 |
| JB GPT-5.4 binary | D | 84 | 35.43 | [5.62, 17.70] | 2.30 | [0.36, 1.15] | 0 |
| JB GPT-5.4 binary | D | 95 | 20.44 | [2.79, 8.16] | 1.33 | [0.18, 0.53] | 0 |
| JB GPT-5.4 binary | U | 5 | -0.19 | [-1.46, -0.32] | -0.01 | [-0.10, -0.02] | 0 |
| JB GPT-5.4 binary | U | 20 | 0.97 | [-3.42, 0.53] | 0.06 | [-0.22, 0.03] | 58 |
| JB GPT-5.4 binary | U | 50 | 10.13 | [0.52, 8.09] | 0.66 | [0.03, 0.53] | 0 |
| JB GPT-5.4 binary | U | 84 | 22.97 | [4.61, 13.08] | 1.49 | [0.30, 0.85] | 0 |
| JB GPT-5.4 binary | U | 95 | 15.66 | [2.37, 6.55] | 1.02 | [0.15, 0.43] | 0 |
| JB mini full | D | 5 | -1.00 | [-3.00, 0.00] | -0.06 | [-0.19, 0.00] | 0 |
| JB mini full | D | 20 | -11.51 | [-23.73, -11.42] | -0.75 | [-1.54, -0.74] | 0 |
| JB mini full | D | 50 | 42.29 | [-8.59, 8.93] | 2.75 | [-0.56, 0.58] | 45 |
| JB mini full | D | 84 | 97.78 | [27.51, 44.76] | 6.35 | [1.79, 2.91] | 0 |
| JB mini full | D | 95 | 61.97 | [15.66, 24.85] | 4.03 | [1.02, 1.61] | 0 |
| JB mini full | U | 5 | -3.87 | [-9.25, -7.07] | -0.25 | [-0.60, -0.46] | 0 |
| JB mini full | U | 20 | -1.89 | [-21.27, -13.60] | -0.12 | [-1.38, -0.88] | 0 |
| JB mini full | U | 50 | 37.13 | [-5.17, 9.89] | 2.41 | [-0.34, 0.64] | 16 |
| JB mini full | U | 84 | 87.48 | [25.17, 40.36] | 5.68 | [1.64, 2.62] | 0 |
| JB mini full | U | 95 | 55.57 | [14.28, 22.56] | 3.61 | [0.93, 1.47] | 0 |
| JB mini binary | D | 5 | 1.00 | [-4.00, 2.00] | 0.06 | [-0.26, 0.13] | 44 |
| JB mini binary | D | 20 | -0.38 | [-9.05, 2.06] | -0.03 | [-0.59, 0.13] | 4 |
| JB mini binary | D | 50 | 7.96 | [-1.99, 5.40] | 0.52 | [-0.13, 0.35] | 15 |
| JB mini binary | D | 84 | 15.75 | [1.86, 8.05] | 1.02 | [0.12, 0.52] | 0 |
| JB mini binary | D | 95 | 10.16 | [1.01, 4.16] | 0.66 | [0.07, 0.27] | 0 |
| JB mini binary | U | 5 | -0.26 | [-1.33, -0.07] | -0.02 | [-0.09, -0.00] | 0 |
| JB mini binary | U | 20 | 0.43 | [-2.72, 1.32] | 0.03 | [-0.18, 0.09] | 52 |
| JB mini binary | U | 50 | 6.56 | [-0.80, 6.29] | 0.43 | [-0.05, 0.41] | 7 |
| JB mini binary | U | 84 | 15.06 | [1.51, 7.89] | 0.98 | [0.10, 0.51] | 0 |
| JB mini binary | U | 95 | 10.21 | [0.78, 3.86] | 0.66 | [0.05, 0.25] | 0 |

# C Certification and repair under different allocations

Let $E_0 = \sum_i \mathbf{1}\{\widehat{Z}_i^0 \neq Z_i\}$. For a query set $Q$, write $C_W(Q)$ and $C_R(Q)$ for the numbers of certified units that were initially wrong and initially right. Then

$$C(Q) = C_W(Q) + C_R(Q), \tag{20}$$
$$E_G(Q) = E_0 - C_W(Q), \tag{21}$$
$$E_E(Q) = E_0 - C_W(Q) - D(Q) + A(Q). \tag{22}$$

Every certified initially wrong unit is repaired under either rule. The only further eager repairs are the uncertified false-FAIL repairs counted by $D$, and its only new errors are those counted by $A$. This proves the last two identities directly from the initial classes.

For two allocations applied to the same frame, define every signed difference as mixed allocation minus uniform criterion sampling. Pairing within a query-order replicate yields

$$\Delta C = \Delta C_W + \Delta C_R, \qquad \Delta E_E = -\Delta C_W - \Delta D + \Delta A, \qquad \Delta E_G = -\Delta C_W. \tag{23}$$

We calculate these differences before averaging, preserving the additive identities in Proposition 3. To describe where additional certificates go, we also report the composition of the *mixed-only* certified set, separately from the uniform-only set. This gross composition remains meaningful when net certification gain is zero or negative. More certificates on initially correct units can provide useful confirmation without repairing an error; it is not by itself evidence that an allocation is harmful.

The fixed mixed design spends up to $\lfloor m/2 \rfloor$ labels on judge-first serial short circuit, then draws a fresh uniform sample without replacement from the remaining criterion slots. If short circuit completes before its phase cap, the unused budget transfers to the random phase. All revealed labels count toward the common actual total $m$. This contrasts acquisition while the matched-query comparison in Appendix A.3 contrasts updating. Neither design uses hidden initial-error classes to choose queries.

# D Finite-population estimation and intervals

## D.1 An adaptive first phase followed by fresh sampling

Set $a_u = \mathbf{1}\{P_u = Y_u\}$. Let the first-phase transcript reveal a set $C$ of $c$ criterion labels with known correct count $t_C = \sum_{u \in C} a_u$. Condition on that complete transcript. The remaining $R = M - c$ slots form a fixed population. Draw an independent simple random sample $S$ of size $n$ without replacement from it, and observe $x = \sum_{u \in S} a_u$. For $n > 0$,

$$\widehat{a} = \frac{t_C + Rx/n}{M}, \qquad \mathbb{E}[\widehat{a} \mid C, \text{first-phase observations}] = a. \tag{24}$$

The expectation follows from inclusion probability $n/R$ for every remaining slot. Conditioning on the observed history accommodates the adaptive first phase: the second-phase estimate expands only the unobserved remainder. When $R = 0$, the reference agreement is known exactly. When $R > 0$ but $n = 0$, we report logical bounds $[t_C/M, (t_C + R)/M]$ and no design-unbiased point estimate from this formula.

Let $S_R^2$ be the finite-population variance of the remainder's agreement bits, with denominator $R - 1$. For $R > 1$ and $n > 0$,

$$\mathrm{Var}(\widehat{a} \mid \text{first phase}) = \frac{R^2}{M^2}\left(1 - \frac{n}{R}\right)\frac{S_R^2}{n}. \tag{25}$$

The factor $1 - n/R$ is the finite-population correction. Census remainders have zero variance. Where a normal approximation is reported, $n \geq 2$ permits the sample variance $s^2 = x(n-x)/[n(n-1)]$ to replace $S_R^2$. That approximation is evaluated separately from the guaranteed intervals below.

## D.2 Inclusive hypergeometric inversion

For unknown remainder success count $K$, the sample count has probability

$$\Pr_K(X = j) = \frac{\binom{K}{j}\binom{R-K}{n-j}}{\binom{R}{n}}. \tag{26}$$

Retain integer counts consistent with the sample and both inclusive tails:

$$\mathcal{K}_\alpha(x) = \left\{K \in \{x, \dots, R-n+x\} : \Pr_K(X \geq x) \geq \frac{\alpha}{2}, \quad \Pr_K(X \leq x) \geq \frac{\alpha}{2}\right\}. \tag{27}$$

Let $K_L$ and $K_U$ be its minimum and maximum. The microagreement interval is $[(t_C + K_L)/M, (t_C + K_U)/M]$. For every fixed true $K$, rejection by either tail occurs with probability at most $\alpha/2$; a union bound gives coverage at least $1-\alpha$. Discreteness can make it conservative. This is a conditional, fixed-budget randomization guarantee with the reference held fixed, not simultaneous coverage of every budget or validity under optional stopping. At $n = 0$ the count range is $[0, R]$, and at $n = R$ it is $[x, x]$.

The implementation compares ordinary tails numerically, but recomputes near-threshold decisions using integer binomial coefficients and rational $\alpha/2$. A tolerance triggers exact calculation rather than accepting an endpoint. For example, with $R = 16, n = 2, x = 0, \alpha = 0.05$, $K = 13$ satisfies $\Pr(X = 0) = \binom{3}{2}/\binom{16}{2} = 0.025$ exactly and must remain in the interval $[0, 13]$. This resolves the floating-point boundary case without altering the mathematical acceptance rule.

## D.3 Prediction strata and a charged pilot

Prediction strata are the nonempty sets with $P_u = 0$ or $P_u = 1$, known before reference acquisition. Proportional allocation reserves one draw per nonempty stratum when affordable, distributes remaining draws proportional to stratum size, respects capacities, and resolves integer remainders by descending fractional part with prediction-zero first on ties. Otherwise it falls back to uniform sampling.

For the charged-pilot design, the initial allocation is $\min(2, N_h)$ labels in stratum $h$. The design reserves at least one later draw in each stratum not exhausted by that initial allocation. If the cap cannot fund both stages, it falls back to uniform sampling. Otherwise the pilot target is the larger of this minimum and $\lfloor 0.2m \rfloor$, capped to preserve the later draws. Additional pilot slots are allocated by stratum size, leaving an unobserved slot in each initially noncensused stratum. After pilot counts $p_h$ and observed agreement totals $t_h$, define

$$\widetilde{a}_h = \frac{t_h + 1/2}{p_h + 1}, \qquad R_h = N_h - p_h, \qquad w_h = R_h\sqrt{\widetilde{a}_h(1 - \widetilde{a}_h)}.$$

The remaining budget is allocated by weights $w_h$, with a minimum of one draw from each nonempty remainder and its capacity as an upper bound. The same capacity-constrained proportional allocation and largest-remainder rule determine integer counts. Pilot and remainder samples use separate random-priority streams. All pilot labels are charged to the budget.

Conditional on the pilot, let the remainder sample in stratum $h$ have size $n_h$ and agreement total $x_h$. Then

$$\widehat{a}_{\text{strat}} = \frac{\sum_h t_h + \sum_{h:R_h>0} R_h x_h/n_h}{M} \tag{28}$$

is conditionally unbiased. If $H$ strata are not censused, we obtain a count interval in each at error level $\alpha/H$, sum their lower and upper bounds and add the known pilot total. The union bound proves conditional coverage at least $1 - \alpha$ for the overall mean; independence between stratum interval events is not required. Censused strata contribute their exact counts. The accompanying normal approximation sums the variances in (25), stratum by stratum, and clips endpoints to $[0, 1]$. It is unavailable when a noncensused stratum has fewer than two sampled slots. These design-based controls are not an implementation of StratPPI.

The archived simulation tables retain bias, empirical RMSE, observed coverage and interval width over 31 query-order replicates at each of six shares $0.006, 0.05, 0.20, 0.50, 0.60, 1$. Their empirical RMSE is $[31^{-1}\sum_{s=1}^{31}(\widehat{a}_s - a)^2]^{1/2}$. We use it as a simulation diagnostic. The main precision comparison at the 20% cap uses design risk as follows. Coverage guarantees continue to follow from the sampling design and tail inversion.

## D.4 Design risk at the 20% criterion cap

For a binary population of size $N$ with $K$ successes, an SRS of size $n$ has sample-mean variance

$$v(N,K,n) = \frac{(N-n)K(N-K)}{nN^2(N-1)}. \tag{29}$$

Set $v = 0$ for a census, including an empty remainder. For fixed stratum sample sizes, the overall variance is $\sum_h (N_h/M)^2 v(N_h, K_h, n_h)$, using the actual integer allocations. The estimators are unbiased, so design RMSE is the square root of variance.

For charged-pilot Neyman, let $a_h$ be the predetermined pilot size and $T_h$ its agreement count. Independently across strata,

$$\Pr(T_h = t) = \frac{\binom{K_h}{t}\binom{N_h - K_h}{a_h - t}}{\binom{N_h}{a_h}}.$$

Write $R_h = N_h - a_h$ and let $n_h(\mathbf{t})$ be the integer remainder allocation produced by the smoothed pilot rates above. Conditional on the pilot, (28) has expectation $a$, the fixed-frame microagreement. Thus the total-variance identity gives

$$\mathrm{Var}(\widehat{a}_{\mathrm{strat}}) = \sum_{\mathbf{t}} \Pr(\mathbf{T} = \mathbf{t}) \sum_h \left(\frac{R_h}{M}\right)^2 v\big(R_h, K_h - t_h, n_h(\mathbf{t})\big). \tag{30}$$

We evaluate the complete finite sum numerically, without Monte Carlo sampling of pilot counts. Across the nine cells this includes 564,528 count combinations. The calculation describes the implemented conditional estimator: pilot observations enter as known totals, while only the fresh sample estimates each remainder.

For the mix, first-stage set $C_s$ leaves population size $R_s$ and agreement count $K_s$; the fresh SRS has size $n_s$. Its exact conditional variance is $V_s = (R_s/M)^2 v(R_s, K_s, n_s)$. We integrate this second-stage randomness analytically and average over $S = 31$ first-stage realizations:

$$\widehat{\mathrm{RMSE}}_{\mathrm{mix}} = \sqrt{\overline{V}}, \qquad \widehat{\mathrm{MCSE}} = \frac{s_V}{2\sqrt{S}\sqrt{\overline{V}}}, \quad \overline{V} = S^{-1}\sum_s V_s. \tag{31}$$

Here $s_V$ is the sample standard deviation of the conditional variances. The MCSE uses the delta method and measures remaining first-stage Monte Carlo error. This is an estimate of full-design RMSE, rather than an exact integration over every possible first-stage transcript. Table 7 reports the values; the stored empirical RMSEs use different, noisier summaries of the same sampling designs.

**Table 7:** Criterion microagreement RMSE at the 20% cap, in percentage points. SRS, proportional strata and pilot Neyman use exact design variances. Mix uses the mean of 31 exact conditional variances; its Monte Carlo SE is reported separately.

| Cell | SRS | Proportional | Pilot Neyman | Mix estimate | Mix MCSE |
|---|---|---|---|---|---|
| DR Gemini | 1.306 | 1.279 | 1.330 | 1.813 | 0.00158 |
| DR Qwen | 1.898 | 1.805 | 1.881 | 2.614 | 0.00087 |
| DR GPT-5.4 low | 1.325 | 1.299 | 1.369 | 1.824 | 0.00162 |
| AC Qwen | 2.244 | 2.208 | 2.407 | 2.985 | 0.00196 |
| AC DeepSeek | 1.836 | 1.769 | 1.857 | 2.494 | 0.00301 |
| JB full | 0.558 | 0.542 | 0.589 | 0.744 | 0.00013 |
| JB mini full | 0.609 | 0.594 | 0.650 | 0.816 | 0.00008 |
| JB binary-only | 0.550 | 0.549 | 0.602 | 0.745 | 0.00013 |
| JB mini binary-only | 0.623 | 0.622 | 0.681 | 0.846 | 0.00008 |

## D.5 Whole-task sampling

For estimating $\theta$, the separate whole-task panel fixes the number of sampled evaluation units in advance and draws them uniformly without replacement. Fully revealing an included unit and short-circuiting after its first

reference failure yield the same certified $Z_i$. Their sample means and finite-population task-count intervals are consequently identical, although their criterion-query costs differ. Here sample size is fixed and label cost is random. This experiment is distinct from stopping task sampling at a fixed label cap, which would generally alter inclusion probabilities and require its own inference.

# E Alternative references and task composition

## E.1 Matched reference panels

The matched frame contains 206 outputs with at least two distinct raters. The corresponding 431 annotations are distinguished from the original 1,539 annotation units. Outputs are sorted by base-task and output identity. For each output, distinct-rater annotations are sorted by the SHA256 digest of the literal string `reference-panel-v1|output|rater|annotation`; the first two complete rubric vectors form panels A and B. A separate ordering with salt `judge-anchor-v1` chooses the annotation supplying the saved primary and auxiliary predictions. The selection rule does not inspect label values or agreement. Rubric-slot identities are checked before alignment, and the same selections are used for both targets and both judges. A and B thus refer to symmetric selections across outputs, not globally ranked raters or a chronological improvement in reference quality.

Crossed comparisons distinguish the reference used for acquisition from the reference used for reading and evaluating the resulting query set. For acquisition panel $a \in \{A, B\}$, an adaptive policy obtains $Q^{(a)}$ using that panel's answers. For evaluation panel $e \in \{A, B\}$, queried slots are re-read using $Y^{(e)}$, both updates are computed from those answers, and residual disagreement is measured against the complete $Y^{(e)}$. Off-diagonal comparisons therefore describe counterfactual re-reading of fixed queried items. They do not retain panel A answers while silently treating panel B as truth. Nonadaptive policies have identical query sets for either acquisition panel. Adaptive short circuit can change its stopping decisions and actual query count.

The update grid includes uniform criterion, disagreement-then-random and judge-first short circuit at all 101 budget shares and seeds 1–31. The allocation grid compares the mixed and uniform designs at the six shares listed above, with equal actual query counts within each contrast. The four crossed means share outputs and references and are not four independent replications. Moving from the original frame to the matched frame also changes composition, weighting and judge anchoring; only within-frame, fixed-prediction contrasts isolate the reference-panel change.

## E.2 Base-task composition replay

For the two JudgmentBench GPT-5.4 targets and DR Gemini, we generate 399 bootstrap compositions with root seed 20261008. Each replicate draws the original number of base tasks with replacement and retains every output, rater and criterion belonging to a selected task. The two JudgmentBench targets share base-task multiplicities; DR has a separate random stream. Original release-relative unit order is preserved by visiting original units and cloning each according to its base-task multiplicity. Shared base-task/copy identifiers retain dependence within a replicated task. Each composition gets its own $N$, $M$ and recomputed rounded budget caps.

For updating, every composition uses the exact order expectations in Appendix B for both analytical policies over the full 101-point grid. For allocation, each composition reruns the 20% mixed versus uniform contrast using the same fixed 31-seed panel, taking paired differences before their mean. Thus the latter ranges include sensitivity of a fixed finite Monte Carlo panel, rather than separately estimating its Monte Carlo error. Quantiles use linear interpolation. We show pointwise 2.5th and 97.5th percentiles in counts and per 100 evaluation units, and retain each replicate's recomputed peak and its location. The archived approximate simultaneous construction uses the 95th percentile of maximum absolute grid deviations; it is omitted from the main figure because its constant width obscures the known zero effects at the two endpoints.

This replay evaluates sensitivity to empirical task composition. In JudgmentBench, 30 curated base tasks do not establish an exchangeable sample from a new task population. Raters remain fixed, and reference-panel and composition experiments are conducted separately. Their ranges therefore do not estimate joint

uncertainty over tasks and raters or correct selection bias. A pointwise range at the original curve's selected peak is also not a simultaneous guarantee for that curve.

# F Information disclosure and sharp logical endpoints

## F.1 Query-only identification

With only queried labels and no restrictions linking unobserved values, let $n_P = \sum_i C_i^+$ and $n_F = \sum_i C_i^-$. Every uncertified unit has at least one unobserved slot and no observed failure. Setting all its unobserved slots to one makes it PASS; setting any one of them to zero makes it FAIL. Because the unrestricted completions can be chosen independently by unit, the sharp endpoint range is

$$\theta \in \left[\frac{n_P}{N}, 1 - \frac{n_F}{N}\right], \qquad \text{width} = 1 - \frac{n_P + n_F}{N}. \tag{32}$$

Attainable finite-population values have increments $1/N$; the displayed interval is their endpoint hull. Its width is a logical consequence of missing information, not a sampling confidence interval. A random sample can estimate the mean precisely without certifying every population unit.

## F.2 Fixed references with withheld associations

The disclosure experiment keeps the complete observed labels fixed when constructing a release and changes what the reconstruction procedure may see. Visible information includes task slots and lengths, criterion categories, and the primary predictions. D0 supplies the exact number of reference PASS bits within each task-length-by-prediction group. D1 refines those groups by criterion category. D2 additionally restores the actual reference values attached to all slots of selected whole tasks. These are reference-containing count summaries and restored links, not additional label acquisition.

Let $\mathcal{B}_g$ partition the slots into disclosed groups, with reference success count $t_g$. Unknown binary assignments $x_u$ and task conjunctions $z_i$ must satisfy

$$x_u, z_i \in \{0, 1\}, \tag{33}$$

$$z_i \le x_u \qquad (u \in \mathcal{J}_i), \tag{34}$$

$$\sum_{u \in \mathcal{J}_i} x_u - z_i \le k_i - 1 \qquad (i = 1, \ldots, N), \tag{35}$$

$$\sum_{u \in \mathcal{B}_g} x_u = t_g \qquad \text{for every disclosed group}, \tag{36}$$

$$x_u = Y_u \qquad \text{for restored slots}. \tag{37}$$

The two conjunction inequalities enforce $z_i = \prod_{u \in \mathcal{J}_i} x_u$: any zero slot forces $z_i = 0$, and all one slots force $z_i = 1$. Minimizing and maximizing $N^{-1} \sum_i z_i$ gives the sharp finite feasible endpoints. This is a binary mixed-integer program, not a continuous linear-program relaxation. Every feasible binary assignment is an allowed completion and vice versa, proving sharpness of the optima in this information model. Extra disclosures can only remove completions, and restoring all labels recovers the observed score.

The program separates exactly by task length because every disclosed group lies within one length stratum. Each endpoint has an attained integer witness, independently checked equality and conjunction constraints, and a closed solver objective bound. The solver receives only the constructed release; hidden labels are used by the publisher to form summaries and later to check containment and full recovery. The experiment restores nested whole-task prefixes at fractions $0, 0.1, 0.25, 0.5, 0.75, 1$, with $\lfloor fN \rfloor$ tasks, under ten saved random orders per population and root seed 20260907. The four RuVerBench runs are analyzed separately. Disclosure-order variation is not sampling uncertainty.

D0 already determines the task-uniform mean-criteria reference score:

$$\mu = \frac{1}{N} \sum_i \frac{1}{k_i} \sum_{u \in \mathcal{J}_i} Y_u = \frac{1}{N} \sum_g \frac{t_g}{k_g}, \tag{38}$$

where $k_g$ is the common task length in group $g$. It need not determine the nonlinear conjunction. Because the predicted task score is visible, subtracting it from each reference endpoint gives the feasible correction range. Its sign concerns correction of one judge's score on fixed outputs, not ranking two solver models. A disclosed reference task-pass total would also identify the aggregate mean without revealing every criterion link; the studied disclosure ladder is one sufficient-information comparison, not a universally minimal release scheme.

### F.3 A complete two-task example

Take two tasks with two binary slots each. With no queries or extra information, their conjunction mean lies between zero and one. A trusted total of exactly two PASS bits reduces the feasible means to zero or $1/2$: assignments $(1,0),(1,0)$ and $(1,1),(0,0)$ attain the two endpoints. If both first-task slots are then observed as PASS, the trusted total forces both second-task slots to FAIL, identifying the mean as $1/2$. Without that total, the same two observations leave a range $[1/2,1]$. Thus a task can be unqueried yet logically determined by additional information. Conversely, knowing only that exactly one of the two tasks passes identifies the mean as $1/2$ without identifying which task passes. These examples distinguish label acquisition, individual identification and aggregate identification. The real RuVerBench release contains the task links; the experiment investigates what its coarsened summaries support, rather than alleging a missing-link defect in that release.

# G Generation of the RuVerBench judge panel

The five runs were collected during study development. Gemini and Qwen provide a cross-provider DeepResearch pair; Qwen and DeepSeek provide an AgenticCoding pair; GPT-5.4 low adds a third complete DeepResearch judge. API availability and budget influenced these choices. Each included run covers its full retained domain. Earlier partial GPT-5.4 experiments, unsuccessful model-access attempts, and a separate DeepSeek max-thinking diagnostic are outside this panel. Thus the panel supplies retrospective comparisons, rather than a representative sample of judges or an untouched confirmation set.

**Table 8:** Generation settings for the saved RuVerBench predictions. A cap counts completion tokens. An omitted parameter is recorded as omitted rather than assigned an assumed provider default. All five final runs have zero unresolved labels.

| Run | Requested model and controls | Execution and completion cap |
|---|---|---|
| DR Gemini | `gemini-3.1-pro-preview`; low thinking; temperature 0 | Gemini Batch; 2,048 |
| DR Qwen | `qwen-plus`; thinking disabled; temperature 0 | Singapore DashScope Batch; 2,048 |
| AC Qwen | `qwen-plus`; thinking disabled; temperature 0 | Same Batch controls, six synchronous recoveries; 2,048 |
| AC DeepSeek | `deepseek-v4-pro`; thinking parameter omitted; temperature 0 | Synchronous; 2,048 in the initial probe, then 8,192; six full-run retries at 16,384 |
| DR GPT-5.4 | `gpt-5.4-2026-03-05`; low reasoning; temperature omitted | OpenAI Batch chat completions; 1,024 |

The Gemini and Qwen runs were collected on September 1–2, 2026, DeepSeek on September 2, and GPT-5.4 on September 12. The aliases and controls in Table 8 describe the actual requests; they do not establish that a provider will serve identical model weights later. In particular, the DeepSeek run did not request the upstream paper's max-thinking setting, and its server-side default has not been reconstructed.

**Messages and context.** All judges in a domain use the same single-criterion content, adapted from RuVerBench revision `0af0256870d6322490c7ab15eb6370779b898a37`. DeepResearch uses an English system message and a user message containing the question, saved response and one criterion, requesting yes/no with a justification. AgenticCoding uses a Chinese user message containing the tools, trajectory and one check with its original category and identifier. Its rubric asks the judge to consider assistant messages, tool calls and available reasoning text. Released labels never enter the request messages.

For AgenticCoding, preparation selects the longest message sequence among trajectory records containing tools, or among all records if none contains tools. Tools and messages are serialized as newline-separated JSON without ASCII escaping. Tool-message content is limited to 5,000 characters; assistant content and reasoning text are each limited to 50,000. Longer fields preserve the beginning and end around a truncation marker. No additional whole-context character cap is imposed. The public offline module `src/ruverbench_prompt_protocol.py` preserves the exact templates, serialization and parsers. It reconstructed all 2,458 unique frozen message arrays exactly during the release check: 1,615 DR and 843 AC requests.

**Parsing and recovery.** DeepResearch accepts a yes/no prefix, including a JSON-quoted string, or a single JSON object with an explicit `covered`/`answer` value. Positive forms are yes, true, 1 and positive; negative forms are no, false, 0 and negative. AgenticCoding requires exactly one matching check identifier with result success or fail. Missing and malformed responses remain unresolved until technical recovery; they are never assigned guessed labels.

Gemini and DR Qwen each produced 1,615 parsed labels. Six AC Qwen requests failed in Batch and succeeded synchronously with identical content and settings; one required a rate-limit backoff. DeepSeek's 86-criterion cost probe had six truncations at 2,048 tokens, resolved at 8,192; six remaining full-run truncations were resolved at 16,384. Its runner permits up to three attempts for rate-limit or server failures. GPT-5.4 produced 1,615 parsed labels, all with stop finish reasons. Recovery addressed technical failures, not disagreement with a reference. The final AC runs each contain all 843 labels.

# H Additional displays and reproduction

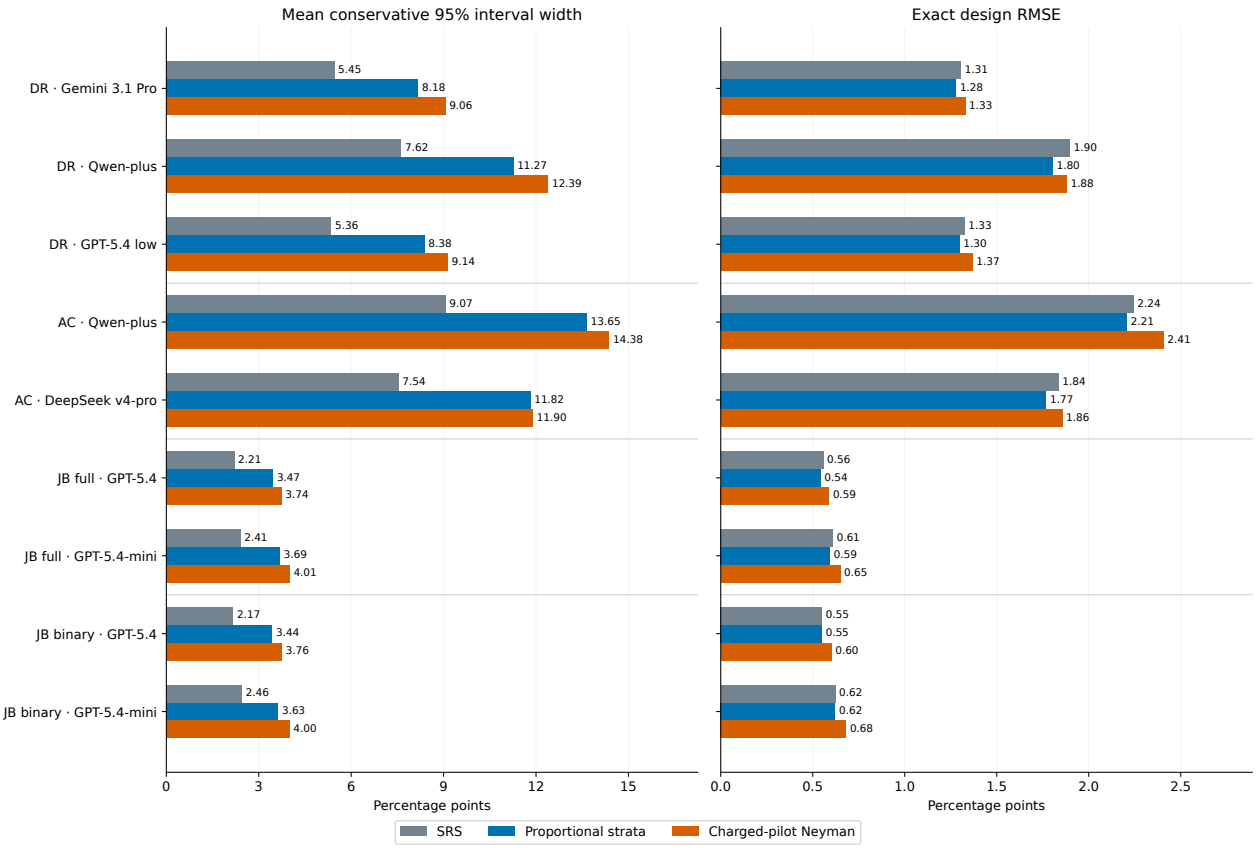



**Figure 3:** Estimation controls at a 20% criterion budget across all nine cells. Mean guaranteed interval widths over 31 saved orders (left) and exact design RMSE (right). Proportional stratification slightly improves RMSE in every cell; the charged-pilot design raises RMSE in eight. The stratified intervals combine count bounds conservatively. Pilot labels count toward the same budget. The coverage guarantee follows from the design.

The released package separates data integrity, scientific replay, and manuscript validation. Inputs are compact numerical projections with task, output, rater, criterion and scoring metadata. Original input provenance and content hashes are in `data/PROVENANCE.json`; result provenance is stored with the archived numerical outputs. The following commands run from the repository root after installing the pinned Python dependencies:

```
python -m src.check_package
python -m pytest -q
python -m src.reproduce --mode full --output artifacts/reproduced
python -m src.verify_release --results artifacts/reproduced
python -m src.reproduce_extensions --output artifacts/reproduced-extensions
python -m src.reproduce_extensions --output artifacts/reproduced-extensions --verify-only
python -m src.verify_manuscript_claims --output artifacts/paper-checks/claims.json
python -m src.build_paper
```

The final command additionally requires Tectonic (tested with version 0.17.0) and writes a rebuilt PDF separately from the published copy. Initial dependency installation can require network access; the scientific replay does not call judge APIs. Rebuilding the PDF can change PDF metadata without changing the scientific results. The published manuscript includes a claim audit and a citation audit; the broader technical reading library remains separate from the selected manuscript bibliography.

Table 9 maps the paper's numerical claims to the result groups. For compactness, paths below are relative to `artifacts/expected/`. The executable claim audit records exact row keys, units, recomputed values and source hashes. It checks selected paper anchors and accounting identities; the complete scientific reproduction commands compare the larger archived grids.

**Table 9:** Numerical source routing for the paper.

| Paper component | Archived source group |
|---|---|
| Frames and initial errors | `conjunction_policy_study/study.json` and input tables |
| Query-order curves | `conjunction_policy_study/budget_curves.csv.gz` |
| Exact expectations and permutations | `conjunction_mechanism/` |
| Paired allocation decomposition | `conjunction_mechanism/` |
| SRS/mix and whole-task estimates | `conjunction_label_value/` |
| Reference and composition sensitivity | `conjunction_robustness/` |
| Stratified estimation controls | `conjunction_robustness/` |
| Disclosure endpoints and witnesses | `information_disclosure_replay.json` |